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CONSTRAINED MOLECULAR GRAPH GENERATION WITH DIFFUSION MODELS

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Abstract

Generative models for molecular graphs have advanced quickly, yet most cannot *guarantee* structural compliance during sampling, which limits their reliability in scientific applications. This thesis aims to enable controllable and provably valid molecular graph generation by integrating strict and interpretable constraints directly into discrete diffusion models. It adapts and extends the CONSTRUCT framework to the QM9 dataset and introduces sampling-time projectors that impose symbolic graph constraints, namely upper bounds on total ring count and on maximum ring length, defined over all simple cycles. The projectors act only during reverse diffusion, as edge-deletion invariant, and reject exactly those edges that would cause a violation, leaving the learned score network unchanged. Across all tested settings, the method achieves perfect satisfaction (100%) of the targeted constraints while maintaining competitive generative quality: RDKit validity, uniqueness, and novelty remain high, and the Fréchet ChemNet Distance stays close to the unconstrained baseline. When constraints are non-binding, the behavior matches the baseline as expected. To make these effects measurable, the analysis distinguishes structural satisfaction at sampling time from chemical validity metrics computed post-hoc and includes diagnostics that track shifts in ring statistics and molecular connectivity under different constraint levels. It also characterizes the trade-offs introduced by tight constraints, such as mild distributional bias and occasional novelty loss. Overall, the results demonstrate that strict and interpretable constraints can be incorporated into discrete diffusion without retraining, enabling controlled molecular generation and providing a general foundation for constraint-aware graph generative models beyond molecules.

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Listing of acronyms

ConStruct	Constraint–projection framework for discrete diffusion
DiGress	Discrete diffusion model for graph generation
EDGE	Degree–guided discrete graph diffusion model
FCD	Fréchet ChemNet Distance
GAN	Generative Adversarial Network
GNN	Graph Neural Network
GraphAF	Autoregressive flow model for graph generation
GraphDF	Flow-based molecular graph generative model
GraphGDP	Continuous–time graph diffusion model
JT-VAE	Junction Tree Variational Autoencoder
logP	Octanol–water partition coefficient
MolGAN	Molecular Generative Adversarial Network
MOSES	Molecular Sets benchmark suite
MoFlow	Flow-based molecular graph generative model
ORGAN	Objective-Reinforced Generative Adversarial Network
QM9	Quantum-chemistry dataset of small organic molecules
RDKit	Open-source cheminformatics toolkit
SELFIES	Self-Referencing Embedded Strings
SMILES	Simplified Molecular Input Line Entry System
VAE	Variational Autoencoder

1

Introduction

Designing novel molecules with desired properties is a fundamental challenge in drug discovery and materials science [3, 4, 5]. Generative models have emerged as a key tool to explore chemical space and propose new molecules *de novo* [6]. Early methods, such as *variational autoencoders (VAEs)* [7, 8], *generative adversarial networks (GANs)* [9, 10], and *autoregressive models* [11, 12], demonstrated that molecules can be generated directly as graphs or sequences. Later, *flow-based models* [13, 14, 15, 16] extended this idea with invertible transformations, providing exact likelihoods and better sample quality. However, all these methods share a fundamental weakness: they tend to generate chemically invalid or unrealistic molecules, often requiring heuristic correction or post-hoc filtering. Fragment-based models such as the Junction Tree VAE [17] improved validity by composing molecules from valid substructures, yet they sacrificed flexibility and offered no systematic way to enforce arbitrary structural rules.

The recent rise of diffusion models has transformed generative modeling, including molecular generation. By progressively adding and removing noise, diffusion models learn to model complex data distributions through iterative denoising [18, 19, 20, 21]. Applied to molecules, *discrete diffusion* has proven particularly effective, most notably in *DiGress* [22], achieved state-of-the-art validity and diversity on QM9 and other benchmarks. Yet, these models still share a key limitation: they capture statistical patterns from data but cannot strictly enforce structural constraints. In molecular design, this means that while the model may learn tendencies such as valence or ring frequency (here, ring refers to a closed loop of covalently bonded atoms forming a cyclic substructure), it cannot guarantee a generated molecule respects those rules.

Recent efforts have tried to address this issue through conditional diffusion or property-guided sampling, where auxiliary networks or gradient-based guidance bias the generation toward molecules with desired numerical properties (e.g., logP, QED, solubility) [23, 24]. In the first case, an additional predictor network estimates target properties during sampling and feeds its signal back to the diffusion process; in the second, the model is steered directly by the gradient of a differentiable property predictor. However, such conditioning remains *soft*: it can encourage but not enforce compliance. Molecules violating desired properties or chemical rules still frequently appear. This highlights a fundamental gap: existing methods can steer distributions, but they still lack mechanisms to guarantee domain-specific constraints.

To bridge this gap, recent research has shifted toward *hard constraint enforcement*, which includes embedding symbolic or structural rules directly into the generative process. Notably, the *ConStruct* framework [2] introduced discrete diffusion with *sampling-time projectors* that correct constraint violations on the fly, successfully enforcing properties such as planarity and acyclicity. Similarly, Ruiz-Botella et al. [25] reformulated diffusion sampling as a constrained optimization process using iterative projections. These works demonstrate that constraint-aware diffusion can achieve both generative flexibility and rule compliance, marking a significant step toward reliable molecular generation.

Nevertheless, current constrained approaches mainly target low-level validity (e.g., valence or connectivity) or simple graph properties, leaving more global and chemically meaningful constraints largely unexplored. Among such global features, **ring structures** stand out as both scientifically important and technically challenging. Rings are widespread in drug-like molecules and strongly influence stability, rigidity, and bioactivity [26, 27]. Controlling the number or size of rings is a routine requirement in medicinal chemistry, which is essential for ensuring synthetic feasibility and optimizing pharmacokinetic properties [28, 29]. Yet, existing generative models, even those with conditional guidance, cannot strictly enforce such limits. They can only *approximate* the desired distributions through training bias.

This thesis directly addresses this gap by extending diffusion-based molecular generation to support hard, user-specified structural constraints on *ring count* and *ring length*. Building upon the *ConStruct* framework, our approach guarantees by design that all generated molecules satisfy these constraints at every step of the diffusion process [2]. This shift, from soft statistical guidance to explicit symbolic enforcement, moves toward generative models that are both expressive and strictly compliant with domain-specific rules.

1.1 PROBLEM DEFINITION AND SCOPE

The central problem of this thesis is **strictly constrained molecular graph generation**: generating novel molecules that *provably satisfy* user-specified structural constraints by construction, while retaining the diversity and realism of unconstrained models. This thesis focuses on small organic molecules represented as undirected graphs (atoms as nodes, bonds as edges), specifically those in the QM9 dataset [30, 31]. Within this setting, we target two constraints:

- **Maximum ring count**: an upper bound on the number of rings in the molecule.
- **Maximum ring length**: an upper bound on the number of atoms in any ring.

Rings are fundamental structural motifs in many organic molecules, including most drug-like compounds. Analyses of approved drugs show that ring systems often form the molecule’s central scaffold (i.e., its core ring-based framework that defines the main molecular topology) [26]. They provide structural rigidity, reduce molecular flexibility, and stabilize the overall 3D geometry. Aromatic rings (with delocalized π -electron systems) and heteroaromatic rings (which include atoms such as nitrogen or oxygen) further modulate polarity and electronic distribution, influencing how the molecule can interact with its environment. In medicinal chemistry, ring statistics are routinely monitored: a high aromatic ring count can reduce solubility and lead to less favorable pharmacokinetic behavior, while very large macrocycles (rings typically containing more than 12 atoms) are harder to synthesize and rarely found in small-molecule drugs [27, 32].

Among the many empirical principles guiding drug design, Lipinski’s Rule of Five [28] remains one of the most influential. It provides simple thresholds on molecular weight, polarity, and hydrogen-bonding capacity that correlate with oral bioavailability, illustrating how rule-based heuristics can effectively constrain chemical space. Although Lipinski’s rule does not explicitly address ring count, it indirectly discourages excessive ring complexity through these limits. Related guidelines, such as lead-likeness and synthetic accessibility, similarly penalize overly complex ring systems [29]. These heuristics inspired our motivation to formalize such domain knowledge within generative models. Controlling ring count and ring size is, therefore, both chemically meaningful and methodologically valuable, making them natural test cases for constraint-aware molecular generation.

From a modeling perspective, ring constraints are *non-local properties*: whether a new bond forms or violates a ring limit depends on the existence of other distant connections in the graph,

not just on local atom valences or bond types. As a result, such constraints cannot be verified or enforced through simple local checks during generation. This makes ring count and ring size particularly challenging test cases for constrained molecular generation frameworks [2].

The proposed approach builds on discrete graph diffusion [22, 21]. The sampling process has been modified so that any edge proposal violating a constraint is rejected by a projection step, ensuring that all intermediate and final graphs remain feasible. This yields a generator that produces molecules satisfying a user-specified maximum ring count and maximum ring length, with no need for post-hoc filtering.

This work does not optimize continuous-valued properties (e.g., solubility or affinity); the focus is on boolean structural constraints. Experiments are limited to QM9-scale molecules for clarity of evaluation; extension to larger molecules is left to future work. We also assume that basic valence validity is handled by the underlying model, or a separate check—valence is not part of our projector [33]. Within these bounds, this work provides a proof of concept: diffusion models can be empowered with hard structural guarantees, demonstrated here on ring-based constraints.

1.2 OBJECTIVES AND RESEARCH QUESTIONS

Given the problem definition above, the objectives of this thesis are as follows:

1. Develop a diffusion-based generative framework for molecular graphs that can incorporate and enforce hard structural constraints during the generation process [2].
2. Evaluate whether enforcing such constraints impacts the quality of the generated molecules in terms of validity, diversity, and distributional similarity to the training data. We seek to achieve high (hopefully perfect) constraint satisfaction without severely degrading other generation metrics.

To drive the research, we pose the following key research questions (RQs):

RQ1: How can structural constraints—specifically, ring-count and ring-size limits—be enforced during the sampling phase of discrete diffusion to guarantee satisfaction by every generated molecule? This question concerns the design of *constraint projectors* that operate at reverse-diffusion time, selectively rejecting edges whose addition would violate a constraint. We investigate how such projectors can be integrated into the sampling process while leaving the diffusion training objective unchanged [2].

RQ2: What is the impact of strict constraint enforcement on the diversity and validity of generated molecules? By addressing this, we aim to understand the trade-offs involved. Does forcing the model to obey ring constraints reduce its ability to explore the full chemical space (leading to loss of diversity)? Does it unintentionally increase the frequency of other types of invalid outputs, or can it maintain chemical validity and novelty at levels comparable to unconstrained models? We will measure standard generation metrics (validity, uniqueness, novelty, as well as coverage/FCD [34]) to answer this.

RQ3: How does the constrained diffusion model compare to baseline approaches in terms of constraint satisfaction and model performance? We will compare our approach to baseline generative models that do not enforce constraints (e.g., an unconstrained diffusion model like *DiGress* [22]). This question addresses whether our method truly adds value by virtually eliminating constraint violations, and at what cost to other performance aspects. A successful outcome would show a 100% constraint satisfaction rate, significantly outperforming baselines, while achieving comparable validity and diversity.

RQ4: Can the approach be generalized to other types of constraints beyond rings? While ring constraints are our immediate focus, we examine conceptually how the method could handle different hard constraints (e.g., composite drug-likeness rules). This is a more open-ended question, addressed in discussion: we analyze the strengths and limitations of our framework in a broader context of constrained generation.

By answering these questions, we aim to thoroughly characterize the proposed method’s capabilities and limitations, thus establishing a foundation for strictly constrained molecular generation using diffusion models.

1.3 CONTRIBUTIONS OF THIS THESIS

This thesis makes the following contributions to the field of molecular generative modeling:

- **Extended *ConStruct* framework for ring constraints:** We introduce a novel extension of the *ConStruct* diffusion framework [2] tailored to enforce ring-structure constraints. In particular, we design new *projector modules* that integrate into the discrete denoising diffusion process [22, 21], actively correcting any violation of a specified ring

count or ring size limit at each reverse diffusion step. To our knowledge, this is the first implementation of hard ring constraints within a generative diffusion model for molecules. Whereas *ConStruct* demonstrated hard constraints for properties like planarity and acyclicity, our work extends the concept to more complex cyclic constraints and shows how to enforce upper-bounded structural features.

- **Impact of constraint enforcement on generation quality:** We investigate how imposing structural constraints influences the core metrics of molecular generation. Beyond standard measures such as validity, uniqueness, novelty, and FCD [34, 29], we introduce a “Constraint Satisfaction Rate” to capture the reliability of constraint adherence. This perspective shifts the evaluation focus from absolute performance toward understanding how hard constraints interact with generative quality—whether they restrict diversity, alter data distribution, or can coexist with realistic molecular variability. This analysis framework highlights the broader implications of integrating strict structural rules into diffusion-based molecular generators.
- **Analysis of strengths and limitations:** We offer a critical analysis of the proposed approach. We explain why the projector-based diffusion is effective for ring constraints (highlighting the edge-absorbing noise and projection strategy that keeps the process in the feasible set), and we discuss its limitations. We also identify scenarios where the current method might struggle (such as extremely strict or unusual constraints not well-represented in training data) and suggest directions for overcoming these issues. This analysis contributes to a deeper understanding of constrained generative modeling, laying the groundwork for extending the approach to other constraints (e.g., incorporating pharmacophoric rules or synthesizability checks in the generative loop) [32, 33].

In summary, the thesis contributes a practically useful and theoretically interesting advancement: a diffusion-based generator that can strictly follow the complex structural rules (demonstrated on ring-based constraints). This advances the state-of-the-art in molecule generation by bridging the gap between flexible deep generative models and rule-based expert systems, achieving the benefits of both. The methods and insights presented may inspire further research into constraint-aware generative models in chemistry and other domains.

1.4 THESIS STRUCTURE

The remainder of this thesis is organized as follows:

- **Chapter 2: Background and Related Work** – We present a unified overview that merges theoretical background with recent advances in molecular graph generation. The

chapter begins with molecular representations and chemical constraints, then surveys major deep generative paradigms for molecules (VAEs, GANs, flows, and diffusion models). Particular emphasis is placed on discrete diffusion for graphs and on frameworks that incorporate symbolic structural constraints, such as *ConStruct* [2].

- **Chapter 3: Proposed Model & Methods** – We present the constrained diffusion framework. This chapter formalizes the problem of constrained graph generation, then describes the architecture and algorithms of the approaches used. We detail the discrete diffusion process on molecular graphs, the design of the ring constraint projector modules, and any modifications to the training procedure required to accommodate the constraints.
- **Chapter 4: Experimental Assessment** – We empirically evaluate our approach on the QM9 dataset [30]. We first define the specific constraint settings tested and describe baseline models for comparison. We then report quantitative results on constraint satisfaction, validity, uniqueness, novelty, and FCD [34] for all models. The results are analyzed to answer the research questions: we examine whether our method achieves the intended strict enforcement and how it compares to baselines in other metrics.
- **Chapter 5: Discussion & Takeaways** – We interpret the findings, discussing what they imply for constrained molecule generation. Limitations of our approach are examined, and we also explore how the method might extend to other constraints. Potential impacts on real-world molecular design workflows are considered.
- **Chapter 6: Conclusion and Future Work** – This final chapter concludes the thesis by summarizing the key achievements and answers to the research questions. We reflect on the broader significance of being able to strictly enforce structural constraints in generative models. Finally, we outline promising avenues for future research, such as scaling up to larger and more complex molecular datasets, combining multiple simultaneous constraints, and applying the framework to other graph generation domains beyond chemistry.

By following this structure, the thesis builds from fundamental concepts to the specific innovations and results, and then to broader implications. The aim is to provide a coherent narrative: starting from why constrained molecular generation matters, through how we achieved it with diffusion models, to what it means for the field going forward. The reader is invited to proceed to Chapter 2 for the detailed literature review that grounds this work in the context of prior research.

2

Background and Related Work

2.1 MOLECULAR REPRESENTATIONS AND STRUCTURAL PROPERTIES

2.1.1 MOLECULES AS GRAPHS

Molecules are most naturally represented as graphs, where **atoms correspond to nodes** and **chemical bonds correspond to edges**. Each node is labeled with an element type (e.g., C, N, O) and may include additional attributes such as formal charge, aromaticity flags, or hybridization state, while edges carry a bond order or discrete bond type (single, double, triple, aromatic) [35]. This graph abstraction captures the combinatorial structure that defines molecular identity and is widely used in cheminformatics and molecular machine learning [36].

Formally, a molecular graph $G = (V, E)$ consists of an atom set V and a bond set $E \subseteq V \times V$, typically represented by an adjacency matrix together with node- and edge-feature tensors. Each edge $(i, j) \in E$ carries a discrete bond type label (commonly single, double, triple, aromatic in standard molecular datasets), so the edge features form a one-hot encoding over this categorical space rather than a binary indicator of connectivity [30, 31, 29]. Such encodings are convenient for algorithmic manipulation and are the standard input to graph neural networks (GNNs) [37, 38]. In practice, libraries such as NETWORKX and RDKit provide utilities for constructing labeled molecular graphs from structural information [39, 33].

For example, the QM9 dataset represents each molecule as a graph with nodes labeled by atom type and edges labeled by bond type, enabling direct use in graph-based generative models [30]. Figure 2.1 illustrates a concrete example of a molecular graph representation, showing how acetic acid can be encoded using adjacency, node feature, and edge feature matrices.

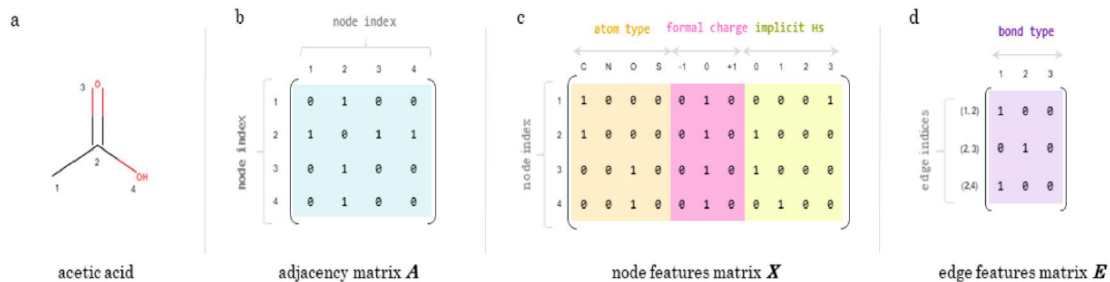


Figure 2.1: Example graph representation of acetic acid, obtained from [36]. (a) Molecular graph with atom indices. (b) Adjacency matrix A encoding bond connectivity under the chosen node ordering. (c) Node-feature matrix X , where each row is a one-hot encoding of atomic attributes such as atom type, formal charge, and the number of implicit hydrogens ("Implicit Hs"). (d) Edge-feature matrix E , where each vector one-hot encodes the bond type (single, double, or triple). Together, A , X , and E illustrate a standard graph-based molecular encoding used in graph neural networks and generative models.

By treating molecules as graphs, we inherently capture the *valence structure* (connectivity) of compounds, which is crucial for generative modeling of chemical structures [40]. Every atom node in a molecular graph typically also has a **fixed degree limit** reflecting its valence (maximum number of bonds): e.g., carbon can form up to 4 bonds, oxygen 2, and so on. Bonds are usually undirected (ignoring geometric directionality), and the graph is thus undirected (or considered as symmetric adjacency matrices in computations). Molecular graphs can include self-loops or multi-edges in theory, but in chemical practice these do not occur (each bond connects two distinct atoms, and multiple bonds are represented via bond order rather than parallel edges). Representing molecules as graphs aligns with established chemical graph theory and allows applying powerful graph algorithms (for substructure search, cycle detection, etc.) to chemical problems [41, 42]. Many molecular generation approaches therefore operate on this graph-based representation of molecules [11, 16, 22, 2].

2.1.2 ALTERNATIVE REPRESENTATIONS

Besides graphs, molecules can be encoded in alternative forms which are often used in generative modeling [36]. A common approach is to represent molecules as linear strings through chemical notation languages.

SMILES (Simplified Molecular Input Line Entry System) is a widely used linear text format introduced by Weininger in 1988 [43]. SMILES encodes a molecular graph as a sequence of characters by listing atoms and using special symbols for bonds, branches, and ring closures (digits in SMILES indicate connections that close cycles). For example, benzene can be written as c1ccccc1 in SMILES notation. SMILES strings are human-readable and compact, making them convenient for storing molecules and using sequence-based models. However, not every arbitrary string corresponds to a valid molecule. SMILES has a context-free grammar, meaning that it must follow a precise set of syntactic rules—such as matching ring indices, properly nested branches, and well-formed atom tokens [44]. If any of these rules are violated, the string becomes invalid and cannot be parsed into a molecular graph. This makes generative modeling challenging: a model producing random SMILES may generate many invalid strings that do not map to real molecules. Moreover, small structural variations can result in disproportionately large changes in their SMILES representations, leading to discontinuities in sequence space (Figure 2.2).

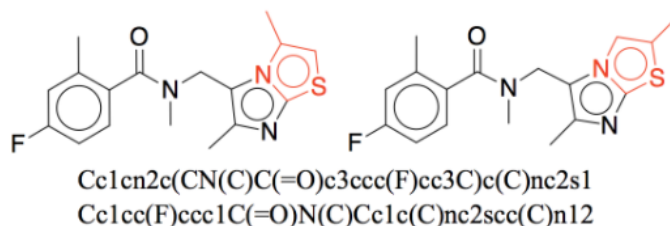


Figure 2.2: Two nearly identical molecules with drastically different canonical SMILES strings (edit distance 22), illustrating the instability of SMILES under small structural perturbations. Adapted from [17].

To address the limitations of SMILES, the **SELFIES** encoding (Self-Referencing Embedded Strings) was proposed as a more robust alternative [45]. SELFIES is a vocabulary of symbols and rules designed such that *any* sequence of SELFIES tokens corresponds to a valid molecule. It achieves this by incorporating semantic rules (like valence satisfaction) into the alphabet and grammar. As a result, generative models that output random SELFIES strings will always produce valid molecules, eliminating the invalidity issue that arises with naive SMILES generation. The trade-off is that SELFIES strings are sometimes longer or less intuitive than SMILES, but they provide a guaranteed mapping onto the chemical space of graphs.

Another important representation is the **3D geometric format**, where a molecule is described by the coordinates of its atoms in three-dimensional space (often along with bond connectivity) [36]. Here, a molecule is not just a graph but a specific spatial arrangement of atoms.

Representations like *Cartesian coordinates* or internal coordinates (bond lengths, angles, etc.) capture the geometry, which is critical for properties like binding affinity or reactivity that depend on shape [36]. 3D generative models (e.g., those for protein-ligand docking or conformation generation) operate on this continuous coordinate space. However, 3D data comes with its own challenges: molecules are flexible (one graph can have many conformations), and generating valid 3D structures requires equivariance to rotations and translations as well as satisfaction of geometric constraints (e.g., reasonable bond lengths and angles) [46, 47].

Each representation has pros and cons. Linear string representations (SMILES/SELFIES) allow one to leverage sequence models (RNNs, Transformers) and avoid dealing explicitly with graph isomorphism or permutation invariance; they have been successful in many early molecular generative works [48, 10, 8]. However, sequence models may struggle to learn the long-range dependencies corresponding to distant parts of a molecule, and syntactic errors can lead to invalid outputs. Graph representations maintain the explicit connectivity and are more aligned with chemistry; generative models on graphs (using GNNs or graph transformations) inherently treat molecules as combinatorial objects [49, 11]. These models must handle the permutation invariance of graph nodes (i.e., the fact that there is no canonical ordering of atoms), which complicates learning but can be handled by appropriate architecture choices (e.g., invariant or equivariant networks) [50, 51]. 3D representations include the most information and are necessary for tasks involving molecular shape, but modeling the joint distribution of connectivity and coordinates is highly complex [47, 52].

In summary, this work (and much of the field) primarily adopts the **graph-based representation** for molecules as it offers a good balance: it encodes exactly the chemical connectivity which defines a molecule, abstracting away conformational variability while capturing the key structural features that generative models need to decide which bonds to form.

2.1.3 STRUCTURAL AND CHEMICAL VALIDITY

Not every graph of atoms and bonds corresponds to a realistic or synthesizable molecule [35]. Ensuring **chemical validity** requires adherence to structural rules that reflect physical and electronic constraints derived from chemistry. Several principles govern whether a molecular graph can represent a plausible compound.

A first requirement is the satisfaction of *valence rules*. Each atom type has characteristic bonding preferences, such that carbon typically forms four bonds, and violations of these limits result in chemically implausible structures. Some representations, such as SELFIES, enforce

correct valence by construction, whereas graph-based generative models must either learn these rules or enforce them explicitly during or after generation [45, 22].

A second aspect concerns *aromaticity*. Aromatic rings such as benzene follow electronic criteria related to conjugation and Hückel’s $4n + 2$ rule [53]. Toolkits like RDKit automatically detect aromaticity and treat aromatic bonds as a separate bond type [33]. Generative models must assign these labels consistently, since improper mixing of aromatic and aliphatic bonds can break chemical validity.

Planarity is another structural consideration [54]. Although many small organic molecules are approximately planar in three-dimensional space, planarity can also be treated as a purely topological constraint. Some generative frameworks impose planarity to discourage unrealistic, overly dense bonding patterns [2]. While planarity is not universally required for chemical correctness, it can serve as a useful inductive bias (a structural assumptions that shape how a model generalizes).

Cycle topology also plays a central role. Rings are fundamental motifs in medicinal chemistry, but drug-like molecules typically contain only a small number of them (often 0–2). Excessive polycyclic complexity correlates with synthetic difficulty and poor pharmacokinetic properties [26, 27]. For these reasons, generative models can benefit from incorporating priors that favor realistic ring counts and ring sizes or limit overly complex cyclic structures.

Finally, molecules of interest are usually assumed to be *connected graphs*. Disconnected samples correspond to multiple fragments or disjoint molecules and are generally treated as invalid unless explicitly permitted by the task. Connectivity may be enforced during sampling or verified afterward using RDKit sanitization [49, 2].

In practice, chemical validity is typically measured as the proportion of generated graphs that pass RDKit’s sanitization routines. Tools like RDKit are essential for validating these conditions post-hoc: given a generated graph, RDKit can attempt to interpret it as a molecule (assign bond orders, check for valence issues, aromaticity, etc.) and report whether the structure is chemically valid [33].

In generative modeling experiments, one typically reports the fraction of generated samples that are valid molecules after this sanitization step (Validity %). High validity is a basic requirement for any practical molecular generator. The aim of incorporating domain constraints into generation is precisely to *increase the validity* of the outputs by ensuring the model only explores the subspace of graph structures that respect fundamental chemical rules [24, 55, 56].

2.1.4 CONSTRAINTS AS SYMBOLIC PRIORS

Incorporating chemical knowledge into generative models can be understood as introducing **symbolic priors**.

A symbolic prior is a rule or set of expert-defined constraints that narrows the space of feasible solutions. In molecular generation, such as including degree–valence limits (e.g., “each atom’s degree must not exceed its valence”). These constraints encode domain knowledge about what constitutes a chemically reasonable structure.

Embedding such priors into the generative process biases sampling toward chemically plausible molecules without requiring the model to infer these restrictions only from data [24, 56, 55]. There are generally two ways to handle constraints in generative modeling: **rule-based enforcement vs. learned (data-driven) enforcement**.

In the first case, constraints are enforced through rule-based mechanisms that intervene directly during generation. For graph-based models, this may involve rejecting candidate edge additions that would violate a specified structural rule. Projector-based diffusion methods fall within this category, ensuring compliance with a constraint by correcting potential violations at every step of the reverse diffusion process [2]. This approach provides hard guarantees by construction.

The second approach relies on learned enforcement, where constraints are only encouraged through training data, regularization, or penalty terms. Such methods often improve chemical validity but cannot ensure perfect adherence, since the model must infer validity rules from data and may still produce invalid structures [22, 23].

Symbolic priors reduce the generative search space, improve interpretability and control, and ensure that critical structural rules cannot be violated. Their main drawbacks are potential reductions in diversity and additional computational cost for constraint checking. In later chapters, we build upon this general principle by incorporating explicit structural constraints into a discrete diffusion model to control molecular topology during generation.

In this work, symbolic constraints are central: they underpin the *ring-count* and *ring-length* projectors we develop to enforce cycle-based structural constraints within a discrete diffusion model. By combining symbolic priors with a learned generative process, we obtain a controllable molecular generator that guarantees structural compliance. A broader comparison between symbolic constraints and learned, data-driven constraints in generative modelling is provided later in Section 2.4.3, where these ideas are revisited in the context of constrained graph diffusion.

2.2 GENERATIVE MODELS FOR MOLECULAR DESIGN

2.2.1 EARLY APPROACHES: RANDOM AND RULE-BASED ENUMERATION

Before the appearance of deep generative models, early computational strategies for exploring chemical space relied on either random brute-force enumeration or manually encoded chemical rules [57].

In **random enumeration**, atoms and bonds are assembled in random combinations under simple constraints to form candidate graphs, which are then filtered for chemical plausibility. For instance, one can sample graphs up to a maximum number of atoms by adding bonds with some probability and subsequently discard graphs that violate basic rules (e.g., valence, connectivity). While conceptually simple, this strategy suffers from a severe combinatorial explosion: the number of possible molecules grows astronomically with size, and most randomly generated graphs are chemically nonsensical [41, 42]. The yield of valid and interesting molecules is therefore extremely low.

To improve upon blind enumeration, **rule-based generative systems** were developed [58]. These rely on human-crafted rules derived from chemistry knowledge, such as known chemical reactions or assembly rules. One classic example is using a library of molecular fragments (common rings, functional groups) and rules to join them (like known chemical reactions or linkers) to construct new molecules. Because every construction step obeys legitimate chemical transformations, the resulting structures are chemically valid by design and can incorporate synthetic feasibility by mirroring real reaction chemistry [59, 60]. The downside is that the chemical space reachable is limited by the predefined rules and fragments. It’s only as creative as the rules set; novel structures outside that template won’t be found.

By the mid-2010s, it became clear that purely rule-based or random methods could not efficiently balance coverage of chemical space with validity and diversity. This motivated the shift toward *data-driven* generative models that learn the implicit “rules of chemistry” from large molecular datasets rather than relying only on hand-designed rules [29, 31]. Nevertheless, modern deep generative models sometimes still incorporate rule-based components (e.g., generating sequences of reaction steps) [17, 48], reflecting a hybridization of symbolic and learned strategies.

2.2.2 VARIATIONAL AUTOENCODERS FOR MOLECULAR GENERATION

Variational autoencoders (VAEs) were among the first deep generative models successfully applied to molecules [61]. An autoencoder consists of an encoder network that maps an input molecule to a latent vector and a decoder that reconstructs the molecule from this latent representation [62]. A *variational* autoencoder augments this with a probabilistic prior on the latent space, enabling generative sampling by drawing latent vectors and decoding them. Applied to molecular data, VAEs learn a continuous latent space of chemical structures, embedding molecules into a real-valued vector space where similar structures lie close together and smooth movements in latent coordinates correspond to smooth changes in molecular structure; this space can be explored through interpolation or optimized with respect to desired properties [48, 63].

Early work, such as the *Junction Tree VAE (JT-VAE)* by Jin *et al.* (2018), took a graph-based approach: each molecule is decomposed into a tree of molecular substructures (e.g., rings) and a VAE is trained to encode and decode this junction tree along with the associated graph [17]. By generating molecules as assemblies of valid fragments, *JT-VAE* ensures chemical validity by construction for decoded structures and supports constrained optimization in latent space (e.g., improving a property while preserving overall scaffold) [17]. This hierarchical representation helps capture both local and global structure, as illustrated in Figure 2.3.

Another influential example is the *Grammar VAE* by Kusner *et al.* (2017), which operates on SMILES strings and integrates the context-free grammar of SMILES into the decoder [8]. By constraining the decoder to output syntactically valid strings, *Grammar VAE* significantly improves the fraction of valid SMILES compared to a naive character-based VAE. However, syntactic correctness does not guarantee chemical plausibility; it is still possible for the decoder to produce syntactically valid but chemically unrealistic strings (like correct grammar but wrong valence), since the grammar alone doesn't enforce all chemical rules.

Graph-based VAEs such as *GraphVAE* (Simonovsky & Komodakis, 2018) attempted to directly encode molecular graphs into a latent space and decode to graphs without an explicit fragment tree [7]. These models must handle permutation invariance of nodes (often by defining a canonical node ordering or using attention mechanisms) and had to deal with the discrete nature of graphs (some used continuous relaxations of adjacency matrices during training). While able to generate small graphs, *GraphVAE* and similar models struggled to consistently ensure validity; many decoded graphs had broken valence or extra disconnected components.

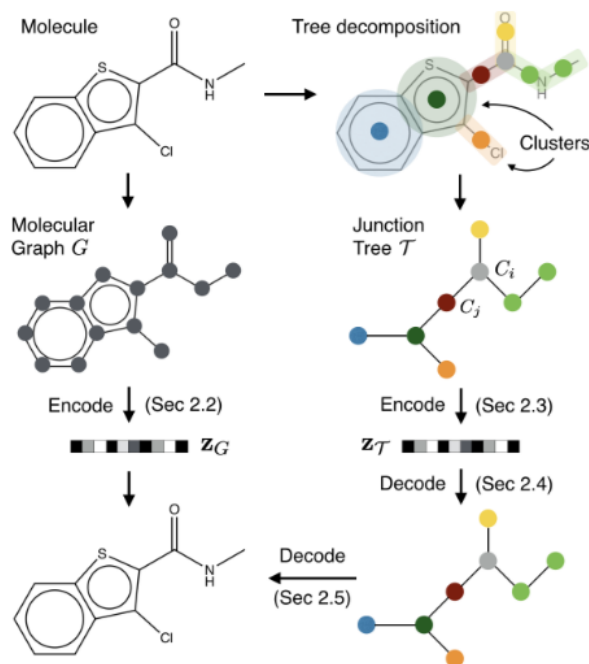


Figure 2.3: A molecular graph is decomposed into a junction tree, where each colored node represents a valid substructure such as a ring or linker. Both the tree and the graph are encoded into latent embeddings (z_T and z_G) and jointly decoded to reconstruct valid molecules. Obtained from [17].

In summary, VAEs introduced the paradigm of **continuous latent representations** for molecules. This was very appealing because it allows one to perform operations like interpolation or gradient-based optimization in latent space to achieve desired molecular properties, and then decode the result to a real molecule. VAEs are generally efficient to train and sample from, but their performance depends heavily on how well the decoder can be structured to produce valid outputs. Without special measures, a vanilla VAE on molecules tends to produce a lot of invalid candidates; but with domain-informed decoders (like the *JT-VAE*’s fragment assembly or *GrammarVAE*’s syntax rules), they can achieve validity rates upwards of Validity $\sim 70\text{--}100\%$ and produce novel, diverse structures [17, 8].

However, the limitation is that VAEs often suffer from **KL-divergence and reconstruction trade-offs**, which means if the model focuses too much on reconstruction, the latent space might not be well-organized, and if it regularizes the latent space too much, it might underfit the data. In practice, careful tuning and architecture choices are needed, and even then, VAEs might not capture the full complexity of molecular distributions (leading to relatively generic molecules if not constrained to be similar to training examples) [29, 31].

2.2.3 GENERATIVE ADVERSARIAL NETWORKS FOR MOLECULAR GENERATION

Generative adversarial networks (GANs) provide an alternative deep generative paradigm in which a generator network G and a discriminator network D engage in an adversarial game [64]. The generator maps random latent vectors to candidate samples (here, molecules), while the discriminator tries to distinguish real molecules from generated ones. The training is a minimax game where G is rewarded for fooling D and D is rewarded for catching G 's fakes [65]. This adversarial setup can, in theory, train G to produce data indistinguishable from real data.

MolGAN (De Cao & Kipf, 2018) is one of the earliest GANs for molecular graphs [9]. The generator directly outputs adjacency and node-feature tensors, which are discretized into a molecular graph, while the discriminator and a property predictor are implemented using graph convolutional networks. *MolGAN* showed that adversarial training can produce reasonable small molecules, but also exhibited typical GAN issues such as mode collapse (limited structural diversity) and unstable training [66]. Because *MolGAN* does not encode explicit chemical validity rules, a non-negligible fraction of generated graphs is invalid and must be filtered post-hoc. The overall architecture is summarized in Figure 2.4.

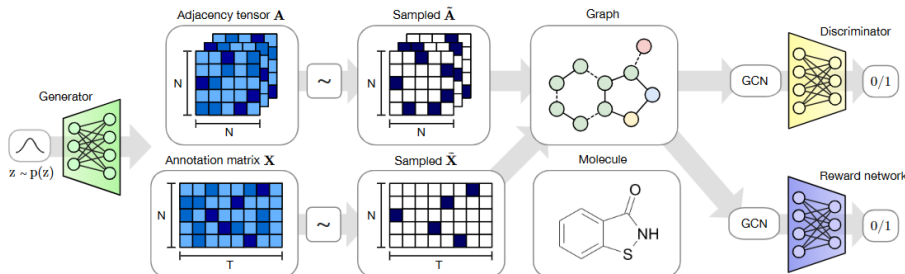


Figure 2.4: The generator samples latent vectors $z \sim p(z)$ and produces an adjacency tensor A and an annotation matrix X , representing molecular connectivity and atom features, respectively. These are discretized into $\tilde{A}$ and $\tilde{X}$ to form an annotated molecular graph. The resulting graph is evaluated by a discriminator and a property-based reward network built on graph convolutional (GCN) layers. Obtained from [9].

In parallel, *ORGAN* (Objective-Reinforced GAN) by Guimarães *et al.* (2017) applied an adversarial framework to SMILES strings, combining GAN training with reinforcement learning [10]. The generator is optimized not only to fool the discriminator but also to maximize task-specific rewards (e.g., predicted activity or drug-likeness), thereby enabling goal-directed sequence generation.

However, as a SMILES-based model, *ORGAN* inherits the fragility of SMILES and does not guarantee 100% validity, instead relying on post-hoc checking and discarding invalid strings. The overall training loop is illustrated in Figure 2.5.

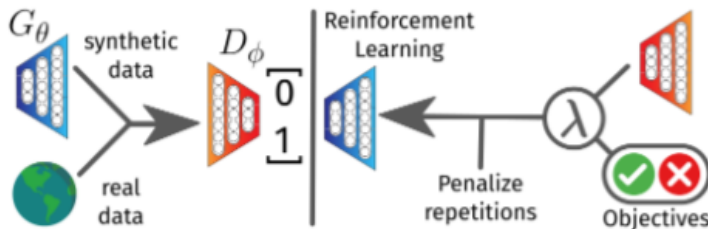


Figure 2.5: The discriminator D_ϕ is trained to distinguish real from generated data, while the generator G_θ is optimized through reinforcement learning to both fool the discriminator and maximize property-based objectives. This reward shaping encourages goal-directed molecule generation but does not guarantee syntactic or chemical validity. Obtained from [10]

GANs can, in principle, generate sharp, high-quality samples by directly matching the data distribution without explicit likelihood modeling. For molecular design, this has the appeal of capturing realistic structures and supporting property-driven objectives. However, in practice, GANs on discrete data suffer from training instability, gradient estimation challenges, and lack a simple latent structure for interpolation or inverse design [65, 66].

Over time, GANs have been somewhat overshadowed by diffusion models for molecule generation, as diffusion methods proved to be easier to scale and yielded higher validity [67]. Nonetheless, the adversarial idea introduced the community to combining generative modeling with property optimization in a single framework, and some later works still draw on adversarial training for fine-tuning molecule generators or in combination with other techniques [68].

2.2.4 NORMALIZING FLOWS FOR MOLECULAR GENERATION

Flow-based models offer yet another approach to generative modeling, one that provides **exact likelihood estimation** and **invertible transformations** [69, 70]. A normalizing flow model transforms a simple base distribution (e.g., a Gaussian) into the target data distribution through a series of invertible, differentiable mappings. By composing these transformations, flows can model complex distributions while allowing exact computation of probabilities (through the change-of-variable formula). In the context of molecules, flow models were designed to map between a continuous latent space and molecular representations [71].

In the molecular context, *GraphNVP* (Madhawa *et al.*, 2019) adapted the RealNVP framework to graphs, defining a reversible mapping from adjacency and node-feature tensors to latent variables [14]. The model uses coupling layers that operate on graph-structured inputs, allowing exact likelihood computation and invertible generation. However, *GraphNVP* must address variable molecular sizes (often via padding to a maximum size) and, like other models, lacks explicit guarantees of chemical validity. *GraphNVP* had moderate success on small datasets (like generating QM9 molecules) but struggled with larger or more complex distributions.

MoFlow (Zang & Wang, 2020) extended flow-based generation with a more chemistry-tailored architecture [15]. It separates atom-type and bond-type channels and designs coupling layers adapted to molecular graphs, achieving higher validity and fidelity on molecule benchmarks. *MoFlow* demonstrates that flows can model complex molecular distributions with high validity on moderate-sized molecules.

Flow-based models combine some of the strengths of VAEs (a structured latent space) and likelihood-based training, but they impose architectural constraints (invertibility, fixed input dimensionality) that are non-trivial for variable-sized, discrete graphs. In practice, flows for molecules often require careful engineering and may still struggle with very large or diverse chemical spaces [16, 13].

2.2.5 LIMITATIONS OF EARLY DEEP GENERATIVE MODELS

While the above generative models -the VAE, GAN, and flow-based paradigms- made significant progress, it is worth highlighting their common challenges, especially in the context of molecular graphs, to motivate newer approaches like diffusion models.

First, all of them struggle with the **discrete, combinatorial nature** of molecular graphs [66]. Small changes in latent space or generator outputs can cause discrete jumps in graph structure or result in invalid molecules. Unlike images, where perturbations typically yield another valid image, molecular graphs occupy a thin, highly constrained manifold in a vast combinatorial space [17, 8, 7]. Models must either enforce constraints explicitly (which complicates architectures and may restrict diversity) or generate freely and discard invalid outputs (which is inefficient and can skew the learned distribution).

Second, these models exhibit issues with **mode coverage**. GANs are particularly prone to mode collapse, where the generator produces a limited subset of scaffolds that fool the discriminator [9, 66].

VAEs, especially if poorly regularized, can suffer from mode averaging: decoding from a latent near the “middle” of many modes may yield unrealistic or chemically implausible hybrids [48]. Flow-based models require sufficiently expressive transformations to capture multimodal distributions; otherwise, they may overrepresent common motifs and underrepresent rare but important structures [14, 15]. Benchmarks such as GuacaMol and MOSES highlight these trade-offs in terms of validity, novelty, and diversity [29, 31].

Third, **chemical validity** remains a persistent challenge. Even with sophisticated decoders (*JT-VAE*, *Grammar VAE*) or carefully engineered flows, a fraction of outputs violate valence or other structural rules [17, 8]. Fragment-based or grammar-based approaches embed some chemistry knowledge into the representation, but this also constrains the expressivity and may limit the reachable chemical space.

These observations motivated the development of generative models that operate through *many small transformation steps*, allowing discrete changes to be handled more gracefully than in one-shot generation and enabling constraint checks throughout the process. This perspective contributed to the rise of **diffusion models**, which iteratively refine a graph starting from noise. Such models naturally create opportunities for enforcing structural constraints during generation, helping to address several of the limitations highlighted above [1, 18].

2.3 DIFFUSION MODELS FOR GRAPH GENERATION

2.3.1 FOUNDATIONS OF DENOISING DIFFUSION MODELS

Denoising diffusion models are a relatively new class of generative models that have proven extremely powerful in domains like image and audio synthesis [1]. The core idea is to define a **noising process** that gradually destroys data, and then learn a **denoising process** that can recover data from noise. The overall diffusion process is illustrated in Figure 2.6.

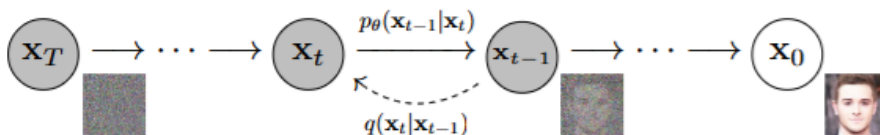


Figure 2.6: The forward process $q(x_t|x_{t-1})$ gradually adds Gaussian noise to data over T steps, while the reverse process $p_\theta(x_{t-1}|x_t)$ learns to denoise step by step, recovering the original sample x_0 . This iterative, Markovian structure forms the foundation of modern diffusion models. Obtained from [1].

In the original formulation for images (continuous data), one adds a small amount of Gaussian noise to an image in T discrete timesteps, eventually turning it into pure noise [1, 72]. The model is trained to invert this process: starting from random noise, predict and remove the noise step by step to yield a coherent image. By optimizing a variational bound, this procedure learns to generate samples that match the training distribution.

Diffusion models can be interpreted as parameterized **Markov chains** (sequences of random transitions where each state depends only on the immediately preceding one) or as stochastic differential equations in the continuous-time limit, building upon earlier work on nonequilibrium thermodynamics [18] and score-based formulations [73].

A crucial challenge arises when extending diffusion to **discrete data** [74]. Gaussian noise cannot be meaningfully added to categorical structures like adjacency matrices, so discrete diffusion models instead define transition processes over categorical states. This requires carefully designed transition kernels and neural architectures that respect the combinatorial structure of graphs.

Unlike one-shot generators such as GANs or VAEs that map a latent vector directly to a sample, diffusion models generate data through a sequence of small, reversible transformations [75]. Each diffusion step modifies the current sample slightly, making it easier to enforce structural constraints or correct errors incrementally. This property makes diffusion especially suitable for discrete domains such as molecular graphs, where validity is fragile and small errors can break chemical correctness.

A key design component is the **forward process schedule**, typically parameterized by a variance β_t that controls how much noise is added at each step, and the **reverse process** that learns to denoise. In continuous domains, the forward noise is Gaussian; however, for **discrete data like graphs**, one cannot meaningfully add Gaussian noise to binary adjacency matrices [75]. Instead, researchers developed *discrete diffusion processes* where node and edge types transition stochastically among categorical states (e.g., edge present/absent or atom type changes). Some methods rely on continuous relaxations of discrete variables [76], while others operate directly in the categorical domain using transition matrices that model probabilities of adding or removing edges.

2.3.2 DISCRETE DIFFUSION PROCESSES FOR GRAPHS

Generalizing diffusion models from Euclidean domains (such as images or audio) to graphs requires redesigning both the forward noising process and the reverse denoiser [77].

The main difficulties include:

- **discreteness**: edges and node types live in categorical spaces,
- **combinatorial explosion**: even small graphs have exponentially many configurations,
- **permutation invariance**: graphs are invariant under node relabeling,
- **structural fragility**: small corruptions (e.g., a single edge addition) can destroy global validity.

Several recent works have developed diffusion processes specifically tailored for graph-structured data [78]. In the forward process of a **discrete graph diffusion**, one usually starts with a real graph and introduces randomness such that after many steps, one reaches a trivial graph (like no edges, or all nodes of a single type). The reverse model then tries to gradually reintroduce the lost information [76, 20].

One prominent method, *DiGress* [22], formulates a discrete denoising diffusion framework for graphs with categorical node and edge attributes. In its forward (noise) process, every node and every (possible) edge independently undergoes a Markovian transition: at each timestep, edges may be added or removed, and node/edge categories may mutate. Over many steps, the graph structure (adjacency) and labels become increasingly corrupted. By the final diffusion step T , the graph is typically highly randomized in both topology and attributes. Learning to reverse this process is carried out by a graph-transformer network: at each denoising step, the model predicts, for every node and every edge, a categorical distribution over possible types. In this way, graph generation becomes equivalent to a sequence of node and edge classification tasks. Figure 2.7 provides a conceptual overview of this process.

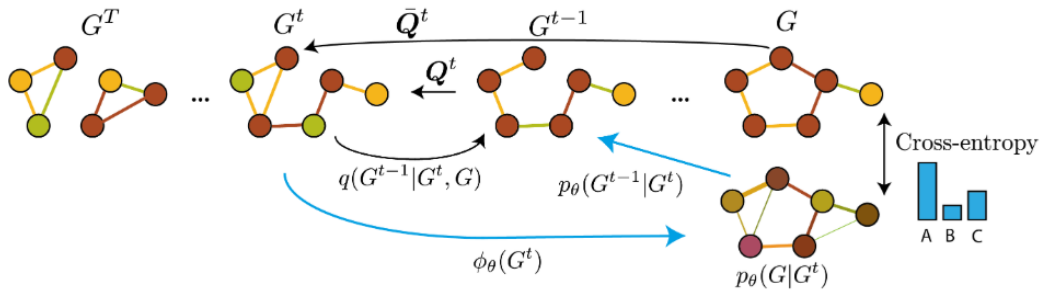


Figure 2.7: Each diffusion step $q(G^t | G^{t-1})$ independently perturbs edges and node/edge labels, while the learned reverse process $p_\theta(G^{t-1} | G^t)$ aims to recover a clean graph via categorical predictions. The model is trained with a cross-entropy loss over node and edge distributions.

A distinctive feature of *DiGress* is its use of a discrete noise model rather than a continuous (e.g., Gaussian) one. Crucially, the diffusion process is designed to converge to the marginal distribution of node and edge types found in the training data, rather than a uniform distribution. This preserves the sparsity and combinatorial nature of graph data throughout the noise trajectory, ensuring that intermediate noisy graphs remain valid (though corrupted) and sparse. As such, structural graph-theoretic quantities (e.g., connectivity, degree, cycles) remain well defined across the diffusion—a property that continuous or uniform noise models often destroy.

Another approach, *GraphGDP* [21], also proposed a discrete diffusion for graphs with a focus on permutation invariance and theoretical grounding. *GraphGDP*’s forward process adds and removes edges in a way that is symmetric and avoids bias due to node ordering. They also considered **continuous-time diffusion** on graphs via jump processes, drawing inspiration from stochastic processes that can handle discrete events [79].

Graph diffusion models typically rely on graph neural networks (GNNs) or attention-based architectures for the reverse process [80]. These must generalize across varying graph sizes, deal with permutation invariance, and capture multi-modal structural distributions [20, 51]. Despite these challenges, discrete diffusion has produced state-of-the-art results on graph generation benchmarks, outperforming VAEs, GANs, and flows in both validity and diversity.

Building on these ideas, the next section examines how diffusion models have been adapted to handle the specific characteristics of **molecular graphs**, including their structural constraints and domain-specific properties.

2.3.3 DIFFUSION MODELS FOR MOLECULAR GRAPHS

Adapting diffusion models to molecular graphs requires incorporating chemical context into the generation process. Molecular graphs are a special subset of general graphs: they have typed nodes (elements) and edges (bond types), and they obey the chemical rules discussed earlier. A generic graph diffusion model might generate any graph, but a molecular diffusion model should ideally generate mostly chemically valid graphs with correct atom types and bonds.

A key development in this direction is the use of diffusion over categorical variables to model both atom and bond types. The molecular variant of *DiGress*, for example, diffuses node labels (elements or “blank” tokens) and edge types (no bond, single, double, triple, aromatic) through a carefully designed transition schedule [22, 2]. This formulation allows the model to denoise toward chemically plausible structures while preserving the inherently discrete nature

of molecular graphs. Despite these advances, unconstrained molecular diffusion still produces high—but not perfect—chemical validity [22].

Diffusion models learn the distribution of valid molecules from data, yet nothing in the standard formulation prevents valence violations or unrealistic bond configurations during sampling. Several works introduce rule-based corrections or masked transitions to discourage invalid updates [23, 24, 56], and others apply lightweight validity checks only at the final denoising step. Such strategies improve validity to some extent but remain heuristic and cannot provide formal guarantees.

Diffusion has also been extended to conditional molecular design [23, 56, 81] and to 3D molecular geometry generation [47, 52, 82]. These approaches highlight the flexibility of diffusion frameworks: properties can be guided, structural motifs conditioned upon, and geometric information incorporated when necessary.

Overall, when it comes to molecular graphs, diffusion models have demonstrated several advantages:

- They often achieve higher validity rates than VAEs/GANs without explicit rules, simply because the distribution of valid molecules is learned and the stepwise denoising tends to favor producing consistent structures [20, 21, 22].
- They can handle multi-modality well: e.g., a molecule dataset might contain very different chemo-types (rings vs acyclic, etc.), and diffusion can generate a wide diversity reflecting that, rather than collapsing to the most common pattern.
- They are relatively straightforward to condition or guide, which is useful for drug design tasks where one might want to enforce a certain substructure or property [23, 56].

In summary, diffusion models have become the state-of-the-art for generative modeling of molecular graphs, outperforming earlier methods in generating valid, novel, and diverse molecules [72, 77, 78]. They serve as the backbone for the **constraint-aware approach**, which will be discussed next: using diffusion as the generative engine, and adding constraint enforcement to guarantee that the samples adhere to desired structural properties throughout the generation process.

2.3.4 MOTIVATION FOR CONSTRAINT-AWARE MOLECULAR DIFFUSION

The limitations above motivate incorporating explicit **symbolic structural constraints** into diffusion-based molecular generators. Because diffusion proceeds in many small steps, it naturally provides points where constraints can be enforced or violations can be projected away

without altering the learned denoising model. This insight underlies recent work on **constraint-aware diffusion** for graphs [2, 25]. By combining a powerful learned reverse model with principled constraint projectors applied at each sampling step, one can guarantee:

- 100% enforcement of structural constraints,
- improved validity and domain fidelity,
- controllability over global molecular structure.

These developments directly motivate the approach used in this thesis: leveraging a *DiGress-style* discrete diffusion model as the generative backbone and integrating symbolic constraint projectors to enforce molecular ring statistics during sampling [22].

2.4 STRUCTURALLY CONSTRAINED GRAPH GENERATION

Despite the progress made with diffusion models for molecules, unconstrained generation can still produce invalid or undesirable structures. In an unconstrained diffusion, the model may occasionally add edges that violate chemical rules (e.g., overbonded atoms or unrealistic polycycles) or produce molecules that are formally valid but far outside drug-like chemical space [22, 20]. For large-scale library generation, even a small proportion of invalid samples becomes problematic, since every invalid molecule represents wasted computation and post-processing effort. Moreover, molecules with extreme ring systems or highly complex scaffolds are typically of little practical value in drug design [26, 27, 28].

These observations motivate the introduction of **structural constraints** that restrict the generative process to a chemically plausible subset of graph space. Examples include:

- *Planarity*: enforcing graph planarity is relevant for certain domains, such as 2D materials or constrained molecular families. Unconstrained generation may occasionally yield non-planar graphs; a planarity constraint explicitly avoids such topologies [2].
- *Acyclicity or tree-likeness*: for polymers or acyclic metabolites, one may require no cycles at all. Without constraints, diffusion can unintentionally form rings; an acyclicity constraint forbids cycle formation [51, 21].
- *Connectivity*: many applications require a single connected component. Diffusion models can generate disconnected graphs; a connectivity constraint removes graphs with multiple fragments [22, 20].

- *Degree (valence) constraints*: atoms must not exceed their typical valence; enforcing this prevents local violations even when global structure appears reasonable [33, 20].

The rationale for enforcing such constraints is twofold. First, **validity and realism**: we want generated graphs to be chemically plausible and aligned with domain knowledge. Second, **efficiency**: constraining the generative process reduces reliance on post-hoc filters that discard invalid samples, which is especially important in goal-directed or iterative optimization settings [24, 23].

Importantly, in a diffusion model, the generative process is a **Markov chain** over many time steps. If the final output must satisfy a structural constraint, it is often beneficial to keep the entire reverse trajectory within the feasible set as well. Allowing intermediate states to violate constraints forces the model to “undo” invalid structure later, which can be difficult and unstable [24, 23, 56, 2]. This motivates **constraint enforcement throughout the trajectory**, not just at the final step.

In summary, structural constraints bring the generative process closer to expert knowledge in chemistry, improving validity, focusing the model on realistic molecules, and reducing downstream filtering. We now review *ConStruct*, a framework that integrates such constraints directly into the graph diffusion process [2].

2.4.1 THE CONSTRUCT FRAMEWORK

ConStruct is a discrete diffusion framework, introduced (2024), that extends the *DiGress* model [22] by adding support for hard structural constraints during generation [2]. At its core, *ConStruct* makes two key modifications to standard graph diffusion:

1. an **edge-absorbing forward process** tailored for constraints, and
2. a **projection operator** applied at each reverse step to enforce the constraint.

The intuition behind *ConStruct* is to restrict the diffusion to operate entirely within the space of graphs that satisfy a given property (e.g., planarity or acyclicity). In a normal diffusion, if you start from a graph that satisfies the property, as noise is added, you could leave that feasible space (e.g., adding a random edge might break planarity). *ConStruct* avoids this by carefully designing the noise addition so that the property is preserved in the forward direction, and by correcting any potential violations in the reverse direction [2].

ConStruct defines a **forward diffusion** that progressively deletes edges until an empty graph is reached. If the initial graph satisfies an edge-deletion invariant constraint, all intermediate

graphs along this forward trajectory automatically satisfy it as well. For example, starting from a planar graph and repeatedly deleting edges can never create non-planar connectivity. Formally, “no edge” acts as an absorbing state in the edge-type space, and the noise schedule is calibrated so that at the final time step all edges have been absorbed [2].

During sampling, the **reverse diffusion** model predicts how to add edges back at each step. However, naive additions could violate the constraint. *ConStruct* addresses this by applying a **projector** $\Pi_{\mathcal{C}}$ after every reverse step, which takes the provisional graph $\hat{G}_{t-1}$ produced by the denoiser (together with the previous state G_t) and maps it back into the feasible set defined by $\mathcal{C}$. The projector removes a minimal set of edges (or modifies them) so that the updated graph again satisfies $\mathcal{C}$. For edge-deletion-invariant constraints, projection can be implemented purely through edge deletions, which fits naturally with the absorbing forward process [2].

What’s remarkable is that *ConStruct* proves that you can integrate this projection into the generative model without biasing the final distribution (under certain conditions). Intuitively, as long as the projection operator is aligned with the absorbing forward process, the correct distribution of constrained graphs can be achieved [2]. This provides a principled way to obtain 100% **constraint satisfaction** while retaining the expressive power of the underlying diffusion model.

A concrete example is the planarity constraint. The forward process deletes edges from a planar graph and thus remains planar at all times. In the reverse process, whenever the model proposes an edge that would break planarity, the projector immediately rejects that edge (or rejects a subset of edges) so that the graph remains planar. Madeira *et al.* demonstrate that this yields planar graph generators with perfect satisfaction of the planarity constraint, in contrast to unconstrained models that occasionally produce non-planar samples [2].

In summary, *ConStruct* turns symbolic structural constraints into *sampling-time projectors* that keep the diffusion trajectory inside the feasible region. This provides a general recipe for combining powerful learned diffusion models with hard guarantees on structural properties. In the remainder of this thesis, we adopt this framework as the basis for our model and instantiate it for molecular graphs with new ring-based projectors (Chapter 3, Sections 3.2–3.3).

2.4.2 RELATED WORK ON CONSTRAINED GRAPH AND MOLECULAR DIFFUSION

ConStruct sits within a broader trend of integrating domain knowledge and constraints into graph and molecular diffusion models. We briefly highlight a few related developments.

EDGE (Efficient Degree-guided Graph Generation) and its improvement *EDGE++* use edge-removal diffusion processes to better match degree distributions and improve scalability on large graphs [83, 84]. Although their primary goal is not hard constraint enforcement, their **edge-absorbing forward process** is conceptually related to *ConStruct*’s design. By biasing diffusion toward sparse graphs and controlling degree statistics, *EDGE* implicitly constrains graph complexity, but it does not employ explicit projectors.

Geometry-Complete Diffusion Models (GCDM) target 3D molecular structure generation and enforce geometric completeness and equivariance during diffusion [82, 47, 52]. These methods encode **geometric constraints**—such as physically plausible distances, and angles—as inductive biases or losses. While they operate in continuous coordinate space rather than purely on 2D molecular graphs, they share the same philosophy: impose domain structure to improve realism and validity.

The Insert-Fill-Halt (IFH) framework introduces a flexible, diffusion-like generative process that interpolates between one-shot and sequential graph generation [85]. *IFH* focuses on controlling the order and granularity of node/edge insertions rather than enforcing explicit structural constraints. In principle, its flexible insertion scheme could be combined with constraint projectors, but this has not yet been explored in detail.

CoCoGraph (Collaborative Constrained Graph diffusion) explicitly targets constrained molecule generation [25]. It integrates chemical constraints such as valence into the diffusion process to guarantee chemical validity and reports improved alignment with real molecular property distributions. Although its mechanism differs from *ConStruct*’s projector-based approach, *CoCoGraph* reinforces the emerging consensus that constraint-aware diffusion is a powerful paradigm for molecular generation.

Overall, these works share a common theme: **the generative process is no longer purely data-driven but is guided by explicit structural or physical principles**. *ConStruct*’s contribution is to provide a general and principled formulation for integrating such constraints via edge-absorbing diffusions and projectors.

2.4.3 SYMBOLIC CONSTRAINTS VS. LEARNED CONSTRAINTS

A central design choice in constrained generative modeling is how much to rely on **symbolic (hard) constraints** versus **learned (soft) constraints** [2, 24, 56]. Symbolic constraints encode expert knowledge as deterministic rules (e.g., valence limits, planarity, cycle-related properties). Their main advantages are:

- **Absolute enforcement:** if the constraint says “planar”, then non-planar molecules simply never appear in the output [2].
- **Interpretability:** constraints correspond to well-understood chemical principles [26, 27, 28].
- **Sample efficiency:** the model does not need to relearn basic chemistry from data; it can focus on more nuanced patterns [33, 22, 20].

Learned constraints, by contrast, incorporate validity or desirability through loss functions, classifiers, or property predictors [23, 24]. For example, one may penalize valence violations, use a learned validity classifier, or bias the model toward high activity or drug-likeness scores. These approaches are more flexible and can capture soft preferences (e.g., “prefer logP in a certain range”) but cannot guarantee strict compliance.

For molecular generation, fundamental chemical rules such as valence and basic ring structure are typically non-negotiable. Relying only on learned constraints risks occasional violations and wastes capacity on modeling invalid regions. For this reason, our perspective is that core structural properties should be enforced **symbolically**, meaning through explicit rule-based checks and corrections rather than learned implicitly, while learning should be used to handle softer, task-specific preferences.

The *ConStruct* framework is naturally aligned with this view: symbolic structural rules are implemented as projectors, and the diffusion model then learns the distribution within the constrained space. This division of labor allows us to combine the reliability of hard constraints with the expressiveness of deep generative models.

Overall, constraint-aware diffusion provides a solid foundation for generative models that are both **chemically reliable** and **task-driven**. The remainder of this thesis builds on these ideas by instantiating concrete projectors for ring-count and ring-length constraints in a *DiGress-style* diffusion model for molecular graphs, and by systematically evaluating their impact on validity, diversity, and distributional alignment.

3

Proposed Model & Methods

3.1 PROBLEM SETUP & NOTATION

We represent a molecule as an undirected labelled graph $G = (V, E)$, where the node set V corresponds to atoms and the edge set $E \subseteq V \times V$ corresponds to chemical bonds. Let $n = |V|$ denote the number of atoms in the molecule.

Each node $v_i \in V$ carries a categorical feature vector describing its atom type, and each potential edge (i, j) carries a categorical bond label. We use one-hot encodings:

$$X \in \{0, 1\}^{n \times d_X}, \quad E \in \{0, 1\}^{n \times n \times d_E},$$

where d_X is the number of atom-type categories (e.g. {C, N, O, F} in QM9), and $d_E = c+1$ is the number of edge-type categories, consisting of c bond types plus a dedicated “no-bond” category representing the absence of a chemical bond between two nodes. Thus, every unordered pair (i, j) takes exactly one of the d_E possible discrete states (bond type or “no bond”), following the discrete edge-type convention used in *DiGress* [22] and adopted in *ConStruct* [2].

We denote by $\mathcal{D}$ the empirical data distribution over molecular graphs (padded to $n_{\max}$ by adding dummy nodes so all graphs have the same fixed size). Every chemically valid molecule corresponds to some graph in this representation, but not every graph corresponds to a chemically valid molecule: an arbitrary graph may violate chemical rules (e.g. valence limits, charge neutrality, forbidden atom–bond combinations).

We therefore distinguish between:

- **Structural validity:** graph-theoretic feasibility (e.g. no duplicate edges, simple undirected graph); and
- **Chemical validity:** satisfaction of domain-specific chemical rules.

Our diffusion model is trained purely on the discrete graph representation (X, E) , where X and E denote the node- and edge-feature tensors described above. Chemical validity is assessed only post-hoc (Section 4.1.2). All projectors introduced in this work operate strictly on the graph topology; enforcement of chemical correctness is deferred to post-generation filtering. Implementation details are discussed in Section 4.2.

3.2 DISCRETE DIFFUSION FOR GRAPHS

Unlike continuous diffusion models that operate in $\mathbb{R}^d$ and gradually add Gaussian noise to continuous vectors [18, 1], we employ **discrete denoising diffusion**, where the “noise” step is a **categorical corruption** applied over a finite set of discrete symbols. All primary generation variables—atom types, bond types, and (optionally) formal charges—are **discrete categorical variables**. The model therefore perturbs *discrete* symbol assignments rather than continuous latent vectors; only the internal representations of the denoising network (e.g., hidden states of the graph transformer) are continuous. This formulation follows the graph discrete diffusion framework of *DiGress* [22], combined with the edge-absorbing extension of *ConStruct* [2], which adapts discrete diffusion to constrained graph domains (see Section 2.3 for continuous diffusion and Section 2.4.1 for a high-level overview of *ConStruct*).

For a summary of continuous diffusion models, see Section 2.3. In the remainder of this section, we describe our discrete generative process—the forward (noising) chain followed by the reverse (denoising) chain—also **illustrated in Fig. 3.1**. The figure presents both the forward and reverse diffusion stages, highlighting how the *ConStruct* model integrates constraint-aware projectors during sampling while maintaining the standard discrete diffusion training objective used in the official *ConStruct* implementation.

3.2.1 FORWARD (“NOISING”) PROCESS

We define a fixed T -step categorical Markov chain

$$q(G_t \mid G_{t-1}), \quad t = 1, \dots, T,$$

which progressively corrupts a molecular graph $G_0 \sim \mathcal{D}$ toward a known prior distribution. This follows the discrete forward process of *DiGress* [22], extended with the *edge-absorbing* formulation of *ConStruct* [2].

We index the nodes in V as $\{1, 2, \dots, n\}$. For an undirected graph we consider unordered edges (i, j) with $1 \leq i < j \leq n$.

The forward process is defined independently for node and edge types through transition matrices:

$$Q_t^X \in \mathbb{R}^{d_X \times d_X}, \quad Q_t^E \in \mathbb{R}^{d_E \times d_E},$$

where

$$q_t(x_{i,t} \mid x_{i,t-1}) = Q_t^X[x_{i,t-1}, x_{i,t}], \quad q_t(e_{ij,t} \mid e_{ij,t-1}) = Q_t^E[e_{ij,t-1}, e_{ij,t}],$$

and one of the d_E edge categories corresponds to the dedicated *no-bond* (no-edge) state.

The cumulative transition over t steps factorises across nodes and edges. For clarity, the edge component can be written explicitly as:

$$q(G_t \mid G_0) = \prod_{1 \leq i < j \leq n} Q_{\leq t}^E[e_{ij,0}, e_{ij,t}], \quad Q_{\leq t}^E = Q_1^E Q_2^E \cdots Q_t^E,$$

and analogously for node types with $Q_{\leq t}^X$. The expression above is the edge-wise part of the full graph transition; together with the node-wise transitions it corresponds to the matrix formulation $q(G_t \mid G_0) = (X_0 Q_{\leq t}^X, E_0 Q_{\leq t}^E)$ used in *ConStruct* [2].

EDGE-ABSORBING NOISE MODEL

To ensure that the forward process is compatible with edge-deletion-invariant constraints, *ConStruct* introduces an *edge-absorbing* noise model [2]. In this model, each edge either stays in its current state or transitions to the no-bond absorbing state. Formally, the edge transition matrix at step t is of the form

$$Q_t^E = \alpha_t^{\text{ABS}} I + (1 - \alpha_t^{\text{ABS}}) \mathbf{1}_c e'_E,$$

where α_t^{ABS} decreases from 1 to 0 according to a schedule, $\mathbf{1}_c$ is a vector of ones over the c non-empty edge types, and $e'_E \in \{0, 1\}^{d_E}$ is the one-hot vector corresponding to the no-bond state. In particular, the no-bond state is absorbing:

$$Q_t^E[\text{no-bond}, \text{no-bond}] = 1, \quad q_t(e_{ij,t} = \text{no-bond}) \rightarrow 1 \text{ as } t \rightarrow T.$$

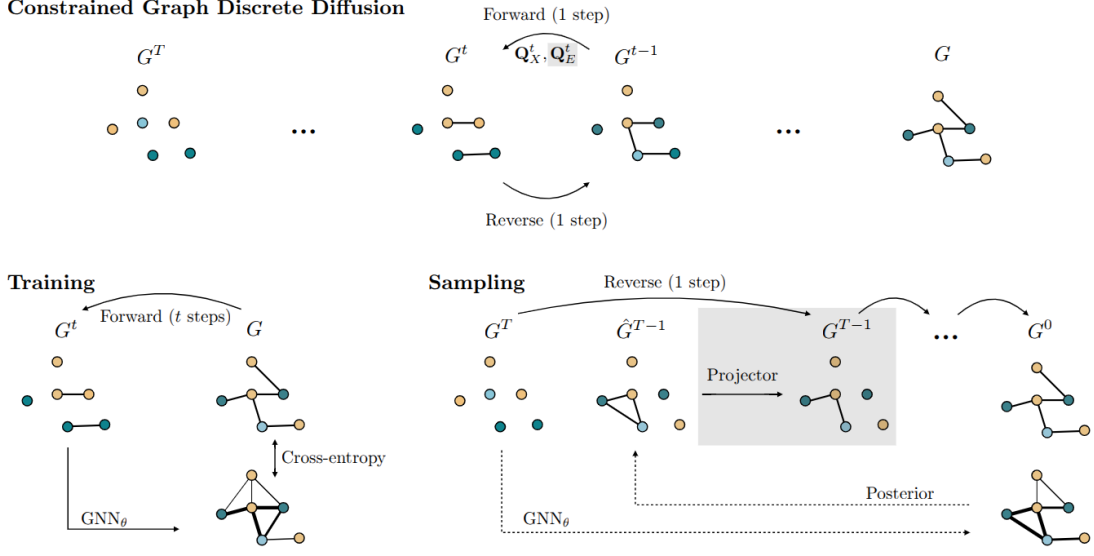


Figure 3.1: Overview of the *ConStruct* framework [2]. The model extends discrete denoising diffusion to graphs and introduces constraint-enforcing projectors at sampling time. Top: forward (noising) and reverse (denoising) chains. Bottom: training (left) and sampling with a projector (right). Obtained from [2].

The prior G_T is therefore an edgeless graph: every edge is in the no-bond absorbing state, while node features are distributed according to a marginal node-noise model as in *ConStruct* [2]. Intuitively, edges are gradually removed as t increases, controlled by a per-step retention parameter $\alpha_t^{\text{ABS}} \in [0, 1]$. Conceptually, α_t^{ABS} plays the same role for discrete edge types as $\alpha_t = 1 - \beta_t$ does in continuous DDPMs [1]: it controls how quickly structure is destroyed.

EDGE-DELETION INVARIANCE

Following *ConStruct* [2], we focus on structural properties that are stable under edge deletion.

Definition 3.2.1 (Edge-deletion invariance [2]). Let $\mathcal{P}$ be a boolean-valued property on graphs. $\mathcal{P}$ is said to be *edge-deletion invariant* if, for any graph $G = (V, E)$ and any subset of edges $\tilde{E} \subseteq E$, the graph $G' = (V, E \setminus \tilde{E})$ also satisfies $\mathcal{P}$ whenever G does:

$$\mathcal{P}(G) = \text{True} \Rightarrow \mathcal{P}(G') = \text{True}, \quad \forall \tilde{E} \subseteq E.$$

Typical examples include planarity, acyclicity, bounded maximum degree, or bounded cycle count. All “at-most” constraints considered in this thesis (e.g. maximum ring count or ring length) are edge-deletion invariant.

Assume that every training graph $G_0 \sim \mathcal{D}$ satisfies an edge-deletion-invariant property $\mathcal{P}$. Under the edge-absorbing forward process, each G_t is obtained from G_0 by deleting a (random) subset of edges and corrupting node labels according to the marginal node noise model. Since edge deletion cannot violate $\mathcal{P}$, every intermediate state G_t also satisfies $\mathcal{P}$:

$$\mathcal{P}(G_0) = \text{True} \implies \mathcal{P}(G_t) = \text{True} \quad \forall t \in \{0, \dots, T\}.$$

Thus, *provided that the data distribution $\mathcal{D}$ lies in the constrained domain*, the forward process never leaves the feasible set of $\mathcal{P}$. This motivates *ConStruct*’s design: constraint preservation can be ensured purely at sampling time by a projector acting on the reverse process, without changing the training objective or forward noise model.

3.2.2 REVERSE (“DENOISING”) PROCESS AND TRAINING OBJECTIVE

We denote by G_t the molecular graph at diffusion step t , with node and edge feature tensors X_t and E_t , respectively. The reverse process aims to invert the forward corruption and gradually reconstruct a molecular graph by learning a parameterised backward Markov chain

$$p_\theta(G_{t-1} \mid G_t), \quad t = T, \dots, 1,$$

which approximates the true posterior induced by the forward process.

As in *ConStruct*, the denoiser is implemented as a Graph Transformer [2] that receives the current noisy graph G_t and auxiliary timestep-dependent features $h_t = h(G_t, t)$. It outputs categorical distributions over discrete variables (nodes and edges, and optionally charges):

$$(\hat{p}_X, \hat{p}_E, \hat{p}_C) = \text{GNN}_\theta(G_t, h_t),$$

where $\hat{p}_X$ and $\hat{p}_E$ are factorised distributions over node and edge types, respectively. In the official *ConStruct* implementation, the reverse transition $p_\theta(G_{t-1} \mid G_t)$ is constructed from these predictions and the known forward transition matrices according to the discrete diffusion formulation in Appendix A of [2].

TRAINING OBJECTIVE

Following *DiGress* [22] and *ConStruct* [2], the model is trained to predict the clean node and edge types of G_0 from the noisy input G_t .

Let X_0, E_0 denote the clean node and edge tensors of the original graph G_0 , and $\hat{p}_X, \hat{p}_E$ the model’s predicted categorical distributions at timestep t . The discrete diffusion loss takes the form

$$\mathcal{L}_{\text{diff}}(\theta) = \mathbb{E}_{G_0 \sim \mathcal{D}, t \sim \text{Unif}\{1, \dots, T\}, G_t \sim q(G_t | G_0)} \left[\text{CE}(\hat{p}_X, X_0) + \lambda \text{CE}(\hat{p}_E, E_0) \right],$$

where $\text{CE}(\cdot, \cdot)$ denotes categorical cross-entropy and $\lambda > 0$ balances node and edge supervision, as in the *ConStruct* training code. This objective corresponds to the standard discrete diffusion training objective used in the *ConStruct* repo: the denoiser learns to reconstruct cleaner graphs from noisier ones by minimising a per-variable classification loss, without explicitly matching the full posterior $q(G_{t-1} \mid G_t, G_0)$. The overall training procedure is summarised in Algorithm 3.1.

We follow this training setup (optimizer, schedules, batch sizing, and early stopping) as in the official *ConStruct* implementation; the implementation details are consolidated in Section 4.2.

Algorithm 3.1 Training for Discrete Graph Diffusion (*ConStruct*-style)

Require: Dataset $\mathcal{D}$ of molecular graphs; timesteps T ; denoiser GNN_θ

- 1: **while** not converged **do**
 - 2: Sample a clean graph $G_0 \sim \mathcal{D}$ and a timestep $t \sim \text{Unif}\{1, \dots, T\}$.
 - 3: Sample $G_t \sim q(G_t \mid G_0)$ using the edge-absorbing discrete noise process
 - 4: Compute auxiliary features $h_t \leftarrow b(G_t, t)$.
 - 5: Predict categorical distributions $(\hat{p}_X, \hat{p}_E, \hat{p}_C) \leftarrow \text{GNN}_\theta(G_t, h_t)$.
 - 6: Compute the loss $\mathcal{L}_{\text{diff}} = \text{CE}(\hat{p}_X, X_0) + \lambda \text{CE}(\hat{p}_E, E_0)$.
 - 7: Update θ by a gradient step on $\mathcal{L}_{\text{diff}}$.
-

3.2.3 SAMPLING WITH PROJECTORS

We denote by P an edge-deletion-invariant property (e.g., bounded ring count, planarity, or connectivity) and by Π_C the corresponding projector that enforces P during sampling, as in *ConStruct* [2]. The node-count prior $\mathcal{P}(|X|)$ specifies the initial number of nodes from which generation starts. Sampling proceeds by iteratively applying the denoiser and projector through all diffusion steps, as summarised in Algorithm 3.2 and shown schematically in Figure 3.2.

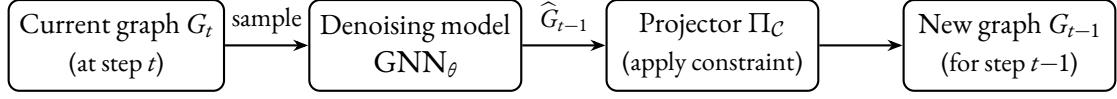


Figure 3.2: Reverse diffusion with a sampling-time projector enforcing a structural constraint, following the *ConStruct* framework [2].

Algorithm 3.2 Sampling with Edge-Deletion-Invariant Projector (*ConStruct*-style)

Require: Timesteps T ; denoiser GNN_θ ; property P ; node-count prior $\mathcal{P}(|X|)$.

- 1: Sample $n \sim \mathcal{P}(|X|)$ and initialise G_T as n isolated nodes (edge-absorbing prior; node features from marginal node distribution).
- 2: Initialise a global blocklist $B \leftarrow \emptyset$ $\triangleright$ edges rejected in any previous step
- 3: **for** $t = T, T-1, \dots, 1$ **do**
- 4: Compute auxiliary features $h_t \leftarrow h(G_t, t)$.
- 5: Predict categorical distributions $(\hat{p}_X, \hat{p}_E, \hat{p}_C) \leftarrow GNN_\theta(G_t, h_t)$.
- 6: Sample a provisional graph $\hat{G}_{t-1}$ from the model’s discrete predictions combined with the forward schedule (as in [2]).
- 7: Apply projector to enforce property P :

$$G_{t-1} \leftarrow \Pi_C(P, \hat{G}_{t-1}, G_t, B)$$

$\triangleright$ Keep only edge additions that preserve property P ; rejected edges are added to B .

- 8: **return** final graph G_0 .
-

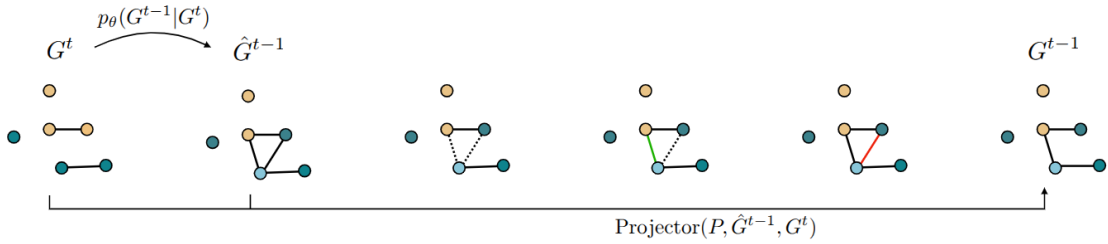


Figure 3.3: Sampling-time projector [2]: after each reverse step, candidate edge insertions proposed by the denoiser (dotted) are filtered by the constraint predicate. Valid insertions are accepted (green), invalid ones (red) are rejected. This guarantees constraint satisfaction without retraining the diffusion model. Adapted from [2].

3.3 SAMPLING-TIME PROJECTORS

Recall from Section 2.4.1 that *ConStruct* enforces edge-deletion-invariant constraints by applying sampling-time projectors during the reverse diffusion process [2]. In this section, we formalise the mechanism shown in Fig. 3.3 and describe the projectors used in our model.

A **projector** is a deterministic mechanism that enforces a structural constraint $\mathcal{C}$ during the reverse diffusion process, correcting forbidden transitions *without modifying* the learned denoising model GNN_θ [2]. At each reverse step $t \rightarrow t-1$, after sampling a provisional graph $\hat{G}_{t-1}$ from the diffusion model, the projection operator

$$G_{t-1} \leftarrow \Pi_{\mathcal{C}}(\hat{G}_{t-1}, G_t)$$

is applied to ensure that the updated graph satisfies $\mathcal{C}$.

Edges whose addition would violate the constraint are deterministically **rejected** rather than deleted or resampled, and the resulting graph G_{t-1} is then passed to the next denoising step. The mechanism is illustrated in Fig. 3.2 and formalized in Algorithm 3.3. As in *ConStruct*, the projector depends explicitly on both the candidate graph $\hat{G}_{t-1}$ and the previous iterate G_t , and it uses a global blocklist of edges that have already been rejected at earlier timesteps.

In our framework, $\mathcal{C}$ always denotes a *graph-structural property* such as ring count, ring length, or planarity. Because edges are only *added*—never deleted—during reverse diffusion (once the edge-absorbing forward process has been applied), the projector acts as a monotone filter that preserves any edge-deletion-invariant property (Section 3.2.1). The learned denoiser remains unaware of this procedure; projection is strictly a *sampling-time* intervention and does not alter the training dynamics.

This modular design allows new constraints to be imposed by substituting the predicate $P(G)$ within $\Pi_{\mathcal{C}}$ without retraining. Multiple constraints can also be composed (e.g., ring count and connectivity), but this is mainly left as future work. The main drawback is potential inefficiency when the denoiser repeatedly proposes invalid edges. To address this, *ConStruct* employs incremental validity checks and maintains a hash-based blocklist of previously rejected edges to avoid redundant evaluations [2]. We adopt exactly the same strategy in our implementation.

Algorithm 3.3 PROJECTOR($P, \widehat{G}_{t-1}, G_t, B$) — Edge-Insertion Filtering

Require: Constraint predicate $P(\cdot)$; provisional graph $\widehat{G}_{t-1}$; current graph G_t ; global blocklist B of rejected edges

```
1:  $G \leftarrow G_t$  ▷ start from current graph (no deletions in reverse)
2:  $C \leftarrow \{\{u, v\} \in E(\widehat{G}_{t-1}) \setminus E(G_t) : u < v\}$  ▷ candidate new edges
3: Randomly permute the candidate set  $C$ .
4: for each edge  $\{u, v\} \in C$  do
5:   if  $\{u, v\} \in B$  then
6:     continue ▷ edge previously rejected at an earlier step
7:   if  $P(G \cup \{\{u, v\}\})$  then
8:      $G \leftarrow G \cup \{\{u, v\}\}$  ▷ accept insertion; property preserved
9:   else
10:     $B \leftarrow B \cup \{\{u, v\}\}$  ▷ reject and block this edge for all future steps
11: return  $G$  ▷ projected graph  $G_{t-1}$  satisfying  $P$ 
```

All constraints used in our experiments are of the “*at-most*” type (e.g., maximum ring count or maximum ring length) and are therefore edge-deletion invariant. During reverse diffusion, this allows the use of a simple *edge-insertion projector*: when adding a candidate edge (i, j) would violate $P(G)$, the insertion is rejected and the edge remains in the no-bond state in the realised graph. Formally, letting G denote the current graph within the projector,

$$G \leftarrow \begin{cases} G \cup \{(i, j)\}, & \text{if } P(G \cup \{(i, j)\}) = \text{True}, \\ G, & \text{otherwise.} \end{cases}$$

This guarantees that every intermediate graph G_t and every final sample G_0 satisfies the constraint $\mathcal{C}$, while the training objective and forward process remain unchanged. This formulation mirrors the constraint-projection mechanism of *ConStruct* [2], enforcing structural properties purely through rejection of invalid edge insertions during sampling.

In what follows, we instantiate this generic projector framework for specific edge-deletion-invariant properties relevant to molecular graphs. We begin with a bound on the total number of simple cycles, then extend the formulation to a maximum cycle-length constraint, and finally demonstrate how the same mechanism recovers the planarity constraint originally proposed in *ConStruct*.

3.3.1 RING COUNT CONSTRAINT ($\leq K$)

The first constraint introduced in this work limits the total number of rings (simple cycles) allowed in the generated molecule. Formally, the property “**ring count** $\leq K$ ” requires that the undirected molecular graph G contain at most K distinct simple cycles.

A *simple cycle* is a closed path that does not revisit any vertex except the starting one. Unlike cycle-basis methods that depend on an arbitrary choice of basis, we count *all unique simple cycles* in the graph. This design choice is motivated by three factors:

- it avoids dependence on the chosen cycle basis and treats isomorphic graphs identically;
- it captures the complete cyclic content of fused or polycyclic systems, including larger “outer” rings omitted by a minimal basis; and
- it defines a property that is **monotone under edge deletion**: removing edges can never introduce new cycles, a requirement for edge-deletion-invariant projectors (Section 3.2.1).

As illustrated below, in Figure 3.4, our ring-count convention treats all simple cycles as distinct rings, so chorded or fused structures contribute both their smaller constituent cycles and any larger outer cycles that enclose them. Thus, counting all simple cycles yields a topologically faithful measure of the cyclic structure, enabling consistent enforcement of both maximum ring count and maximum ring length constraints. Hereafter, we use “*cycles*” (graph-theoretic simple closed paths) and equivalently refer to them in the chemical sense as “*rings*”, interchangeably.

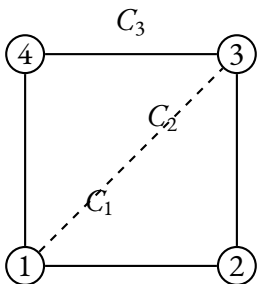


Figure 3.4: Illustration of the *all-simple-cycles* convention used for ring counting. The graph consists of four vertices connected in a 4-cycle (1–2–3–4–1) plus a chord (1–3) (dashed). A typical cycle basis might include only the two triangular cycles $C_1 = (1, 2, 3, 1)$ and $C_2 = (1, 3, 4, 1)$. Under our convention, the larger outer cycle $C_3 = (1, 2, 3, 4, 1)$ is also counted, because it is a simple cycle that does not revisit vertices. Thus this graph has *three* rings, not two, illustrating how fused or chorded systems give rise to additional “outer” rings beyond a minimal cycle basis.

Let $P_{\text{ring-count}, K}(G)$ denote the predicate that is true if and only if the number of simple cycles in G does not exceed K .

During reverse diffusion, edges are only ever added (given the edge-absorbing forward process), so the projector Π_C associated with $P_{\text{ring-count},K}$ ensures that the evolving graph never exceeds K cycles: whenever a newly proposed bond (u, v) would cause the number of simple cycles to surpass K , the edge is deterministically **rejected** and stored in the global blocklist, and the graph remains unchanged. No deletions occur during the reverse process, so once the property holds it remains satisfied throughout sampling.

Algorithm 3.4 formalises the decision rule for verifying whether inserting an edge preserves the ring-count constraint within the generic projector of Algorithm 3.3.

Algorithm 3.4 RINGCOUNTATMOST($G, e = (u, v), K$)

Require: Current graph G satisfying “# simple cycles $\leq K$ ”

```

1: if  $u$  and  $v$  lie in different connected components then
2:   return true ▷ edge cannot create a cycle
3:  $G' \leftarrow G \cup \{(u, v)\}$  ▷ tentative insertion
4: return  $\neg \text{SIMPLECYCLECOUNTEXCEEDS}(G', K)$ 

```

By construction, this design guarantees that every intermediate and final generated molecule satisfies

$$\# \text{ cycles}(G_t) \leq K, \quad \forall t \in \{0, \dots, T\},$$

provided that the initial graph G_T and all projector decisions respect the predicate $P_{\text{ring-count},K}$.

3.3.2 RING LENGTH CONSTRAINT ($\leq L$)

The second structural constraint controls the **maximum ring size** that may appear in a generated molecule. Formally, the property “**maximum ring length** $\leq L$ ” requires that every simple cycle in the molecular graph contain at most L vertices (atoms). Equivalently, the length of the longest simple cycle in G must not exceed L . For instance, setting $L = 6$ permits rings such as benzene (6-cycle) or cyclopentane (5-cycle) but forbids larger macrocycles like cycloheptane. This constraint restricts generation to chemically realistic and synthetically tractable ring sizes.

Let $P_{\text{ring-length},L}(G)$ be the predicate that holds if and only if every simple cycle in G has length at most L . This property is also edge-deletion invariant: deleting edges can only remove cycles or split them, never create longer ones.

During reverse diffusion, the projector Π_C intercepts each proposed edge insertion (u, v) . If inserting this bond would create any simple cycle longer than L atoms, the edge is **rejected**,

added to the global blocklist, and the graph remains unchanged. Because the reverse chain only adds edges and never deletes them, once the property holds it remains satisfied throughout the sampling trajectory. This yields a generation process in which every intermediate graph G_t (and consequently the final molecule) obeys

$$\max_{C \subseteq G_t} |C| \leq L, \quad \forall t \in \{0, \dots, T\}.$$

Algorithm 3.5 outlines the decision rule used to validate each proposed edge insertion under the maximum ring-length constraint, again to be plugged into the generic projector of Algorithm 3.3.

Algorithm 3.5 RINGLENGTHATMOST($G, e = (u, v), L$)

Require: Current graph G satisfies “all simple cycles have length $\leq L$ ”

- 1: **if** u and v lie in different connected components **then**
 - 2: **return true** ▷ edge cannot form a cycle
 - 3: $G' \leftarrow G \cup \{(u, v)\}$ ▷ tentative insertion
 - 4: **return** $\neg \text{MAXCYCLELENGTHEXCEEDS}(G', L)$
-

3.3.3 PLANARITY CONSTRAINT

For completeness and validation, we also implement the **planarity** constraint introduced in *ConStruct* [2]. A graph is *planar* if it can be embedded in the plane without any pair of edges crossing. This property is edge-deletion invariant: removing edges from a planar embedding cannot create crossings. It is trivially satisfied for most small organic molecules—particularly those in the QM9 dataset—whose molecular graphs are almost always planar. Hence, the planarity constraint is included mainly as a baseline and consistency check for the projector mechanism.

Let $P_{\text{planar}}(G)$ be the predicate that holds if G is planar. During reverse diffusion, the planarity projector examines each newly proposed edge insertion (u, v) . If adding this edge would render the tentative graph non-planar, the insertion is deterministically blocked and the edge added to the global blocklist:

$$G_{t-1} \leftarrow \begin{cases} G_t \cup \{(u, v)\}, & \text{if } \text{ISPLANAR}(G_t \cup \{(u, v)\}), \\ G_t, & \text{otherwise.} \end{cases}$$

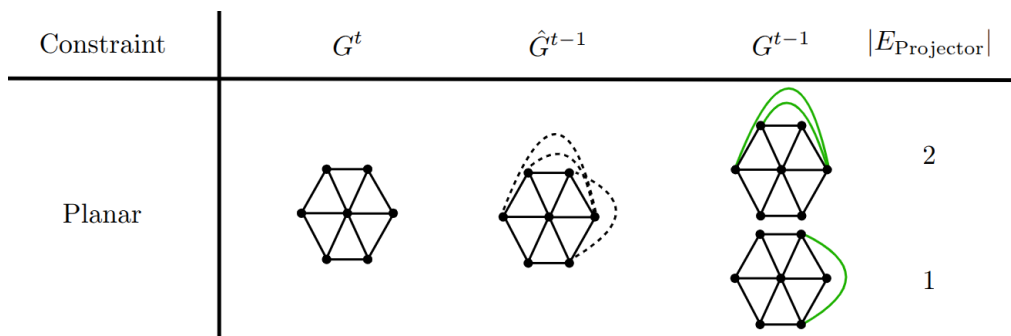


Figure 3.5: Planarity constraint example from *ConStruct* [2]. Proposed insertions that would break planarity (dashed) are rejected by the projector, yielding a planar G^{t-1} . The right column shows the number of accepted edges. Adapted from [2].

The planarity test relies on an efficient linear-time algorithm conceptually following Hopcroft–Tarjan [2, 86]. In practice, we employ the same implementation used in *ConStruct*. Figure 3.5 illustrates how the planarity projector rejects candidate edges that would make the graph non-planar.

Moreover, Algorithm 3.6 summarises the decision rule used to verify whether inserting a new edge preserves planarity.

Algorithm 3.6 PLANARITYCHECK($G, e = (u, v)$)

- | | |
|--------------------------------------|----------------------------------|
| 1: $G' \leftarrow G \cup \{(u, v)\}$ | ▷ tentative insertion |
| 2: return ISPLANAR(G') | ▷ accept iff G' remains planar |
-

3.4 COMPLEXITY AND IMPLEMENTATION EFFICIENCY

This section details the algorithmic strategies and computational efficiency of the projector mechanisms introduced in Section 3.3. The ring-count, ring-length, and planarity projectors must validate candidate edge insertions on the fly to guarantee that every intermediate graph satisfies its corresponding constraint.

Although enumerating all simple cycles is exponential in the worst case, this remains tractable for QM9-scale molecules ($n \leq 9$ heavy atoms). Typical graphs contain only a few dozen cycles, and exact validation completes within milliseconds.

To guarantee constraint satisfaction without false positives, we employ **exact validation** rather than approximate heuristics. For every candidate edge (u, v) , the projector performs a tentative insertion followed by a full structural check:

- **Ring Constraints:** The validation function enumerates all unique simple cycles in the tentative graph $G' = G \cup \{(u, v)\}$ using Johnson’s algorithm [87]. Cycle lists are canonicalised to avoid duplicates arising from graph directionality, ensuring correctness even for fused or overlapping rings. For the ring-count constraint, we verify that the total count does not exceed K ; for the ring-length constraint, we verify that all cycle lengths $|C| \leq L$.
- **Planarity:** We utilize a linear-time $O(|V|)$ planarity test. While our implementation uses the Left-Right Planarity Test [88], it achieves the same optimal asymptotic complexity as the standard Hopcroft–Tarjan algorithm [86], ensuring high efficiency for the sparse graphs found in digital pathology and molecular datasets.

To optimize efficiency, we implement a **caching mechanism** (Algorithm 3.3). All edges rejected by the validator are cached in a global hash-based blocklist B . This prevents redundant evaluations of the same invalid edge in later diffusion steps, significantly reducing the overhead of the exact validation checks. During generation, diffusion steps are executed on the GPU, while projector validations run on the CPU in parallel across the batch. Empirically, the overhead is small—typically a few milliseconds per diffusion step per molecule.

All projectors are purely *structural*, enforcing topological properties such as bounded ring count, ring length, and planarity, without considering chemical feasibility (e.g., valence, aromaticity, or charge). Consequently, chemically invalid but topologically compliant molecules may still appear; these are filtered *post-hoc* during chemical-validity evaluation (Section 4.1.2). Conversely, chemically plausible edges may be rejected if they would violate a structural bound. Future extensions could incorporate chemically aware projectors, such as valence or aromaticity checks, to further align structural and chemical validity.

4

Experimental Assessment

4.1 EXPERIMENTAL SETTING

We adopt a *ConStruct-based* configuration without changes to the training objective or architecture [2]. At evaluation, we draw $N = 10,000$ samples per seed across 5 seeds (50,000 molecules total). Unless stated otherwise, we report mean and standard deviation across seeds for Fréchet ChemNet Distance (FCD) [34], validity, uniqueness, novelty, disconnectedness, and constraint-satisfaction rates. A concise summary of splits, sample counts, and constraint policies is given in Table 4.1; implementation details (training/sampling, software/hardware) are provided in Section 4.2.

Item	Baseline (Unconstrained)	Constrained (Projector-Enabled)
Training / Validation / Test (QM9)	97,732 / 20,042 / 13,054	97,732 / 20,042 / 13,054
Constraints during training	None	None (same trained model)
Constraints during sampling	None	One projector per run: Planarity, Max Ring Count & Ring Length
Projector policy	—	Hard-reject violating edge insertions at each reverse step
Diffusion configuration	Identical across all runs	Identical to baseline
Seeds $\times$ samples/seed	$5 \times 10,000$	$5 \times 10,000$
Total generations (per setting)	50,000	50,000
FCD reference set	QM9 (130,828)	QM9 (130,828)

Table 4.1: Experimental setup summary.

4.1.1 DATASET AND PREPROCESSING (QM9)

We evaluate our models on the QM9 dataset [30], a widely used benchmark for molecular generative modeling and property prediction. QM9 consists of 130,828 small organic molecules composed of carbon (C), nitrogen (N), oxygen (O), and fluorine (F) atoms, each containing up to 9 heavy atoms. The dataset covers a broad range of drug-like and chemically stable small molecules, making it an established testbed for evaluating validity, novelty, and distributional similarity in generative models [7, 13, 22, 21, 83, 2].

For our graph-based experiments, molecules are represented as undirected graphs with atoms as nodes and bonds as edges, over heavy atoms only (C, N, O, F). Hydrogen atoms are handled implicitly by RDKit during sanitization and valence checking; no explicit H nodes are included in the graph. Node features therefore, include only heavy-atom attributes (atom type, formal charge, etc.).

We adopt the standard dataset split of approximately 75%/15%/10% for training, validation, and testing, respectively. No additional filtering or property-based preprocessing is applied beyond RDKit sanitization and canonicalization. Ring statistics used for constraint analysis are computed from all simple cycles in the undirected graph representation, as defined in Section 3.3. A detailed empirical summary of the ring count and ring length distributions within QM9 is presented next in Section 4.3.

4.1.2 EVALUATION METRICS

We evaluate generated sets under standard metrics from de novo molecular generation [31, 29]:

- **Validity (%)** – Fraction of generated graphs that pass RDKit’s chemical sanitization checks (satisfiable valences, correct aromaticity, no chemical violations). While our projectors ensure topological correctness, they do not strictly enforce these chemical rules.
- **Uniqueness (%)** – Fraction of non-duplicate molecules among the valid set.
- **Novelty (%)** – Fraction of unique valid molecules that do not appear in the training set.
- **Fréchet ChemNet Distance (FCD)*** – A distributional distance between generated and reference sets via ChemNet embeddings [34]. Lower is better (0 indicates identical Gaussian summaries):

$$\text{FCD} = \|\mu_{\text{gen}} - \mu_{\text{data}}\|^2 + \text{Tr}(\Sigma_{\text{gen}} + \Sigma_{\text{data}} - 2(\Sigma_{\text{gen}}\Sigma_{\text{data}})^{1/2}),$$

where (μ, Σ) denote mean and covariance of the embeddings.

*<https://github.com/molecularsets/fcd>

- **Disconnected (%)** – Percentage of generated graphs with more than one connected component in the heavy-atom graph (hydrogens implicit).
- **Constraint Satisfaction (%)** – Fraction of molecules that satisfy a specified structural constraint (e.g., ring count $\leq K$, ring length $\leq L$, planarity). With projectors enabled, this is designed to be effectively 100%; we report it for verification.
- **V.U.N. (%)** – A composite score defined as the product of the Valid, Unique, and Novel fractions. It approximates the fraction of molecules that are simultaneously valid, unique, and novel under an independence assumption.

All metrics are computed using **canonical SMILES** strings. We define the denominators hierarchically to ensure consistency: “Unique” is calculated as a fraction of the valid molecules, “Novel” as a fraction of the unique valid molecules, and V.U.N. as a fraction of the total generated samples.

4.2 MODEL TRAINING, SAMPLING, AND IMPLEMENTATION DETAILS

The discrete diffusion model follows the *ConStruct* framework [2], using an *edge-absorbing* forward process in which existing edges independently transition to a special “no-bond” state while node features are perturbed according to empirical marginal distributions. The denoising network is a Graph Transformer with configuration-controlled depth and width, trained using a composite cross-entropy loss over node types and edge types [89].

Formal charges (when present in the dataset) are included as part of the node input features, but they are not predicted as a separate output channel; the model only predicts atom types for nodes and bond existence for edges. Node feature vectors therefore include atom type, formal charge, and other RDKit-derived attributes, whereas the training loss supervises only the atom type and edge-type channels.

Optimization uses AdamW with a constant learning rate of 1×10^{-4} and batch size 128. No learning-rate scheduling, data augmentation, or auxiliary filters are applied. Training continues until early stopping based on validation negative log-likelihood, and final results are reported as the mean across five independent runs with fixed random seeds (0, 1000, 2000, 3000, 4000).

Sampling follows the standard reverse diffusion process p_θ for $T=500$ denoising steps. Each trajectory begins from an edgeless graph G_T with the desired number of nodes. During generation, *constraint projectors* are applied at every reverse step. These projectors reject candidate

edge insertions that would violate a constraint, ensuring that all intermediate and final graphs remain within the feasible set. No deletions or probability re-normalizations beyond this rejection mechanism are performed. For each configuration, 50,000 molecules (five seeds $\times$ 10,000 samples each) are generated to ensure statistically stable metrics. Generated graphs are converted to molecular structures using RDKit [33], including atom typing, implicit hydrogen addition, and sanitization; any molecule failing sanitization is counted as invalid.

All code, trained models, and configuration files are publicly available at <https://github.com/ranaislek/ConStruct-Thesis>. Experiments were conducted on a Linux server with a single NVIDIA RTX A5000 GPU running Python 3.9. Key dependencies include *RDKit 2023.03.2* [33] for molecular handling, *PyTorch Lightning 2.0.4* and *torch-geometric 2.3.1* for model training and data processing, and *NetworkX 2.8.7* [39] for graph-theoretic operations such as cycle detection. The QM9 dataset is accessed through PyTorch Geometric loaders with the standard train/validation/test split described in Section 4.1.1.

4.3 DATASET STATISTICS: RINGS AND PLANARITY

To ground the constrained generation results, we first summarize the ring statistics of QM9. Ring statistics are computed over unique simple cycles in the undirected graph representation (as defined in Section 3.3).

Table 4.2 shows the distribution of ring counts per molecule in the full dataset and each split, which includes train, validation, and test sets. Table 4.3 indicates the distribution of maximum ring length per molecule. Finally, Tables 4.4 and 4.5 give the corresponding cumulative satisfaction rates for ring count and ring length up to various thresholds. The detailed distributions for the full dataset are shown in Figure 4.1 (ring counts) and Figure 4.2 (ring lengths); the cumulative constraint satisfaction rates are plotted in Figure 4.3.

Set (size)	Ring count per molecule k (in %)										
	0 (no rings)	1	2	3	4	5	6	7	8	9	≥ 10
Full (130,828)	10.2325	39.3226	7.3685	24.4527	2.7058	0.0260	5.8413	6.6064	0.3287	0.0015	3.1140
Train (97,732)	10.2924	39.2134	7.2965	24.5037	2.7095	0.0276	5.8538	6.6314	0.3295	0.0020	3.1402
Val (20,042)	10.3233	39.3124	7.4893	24.4686	2.7043	0.0200	5.7829	6.4964	0.3343	0.0000	3.0686
Test (13,054)	9.6446	40.1563	7.7218	24.0463	2.6812	0.0230	5.8373	6.5880	0.3141	0.0000	2.9876

Table 4.2: Distribution of molecules by ring count k . Set sizes are shown in the first column.

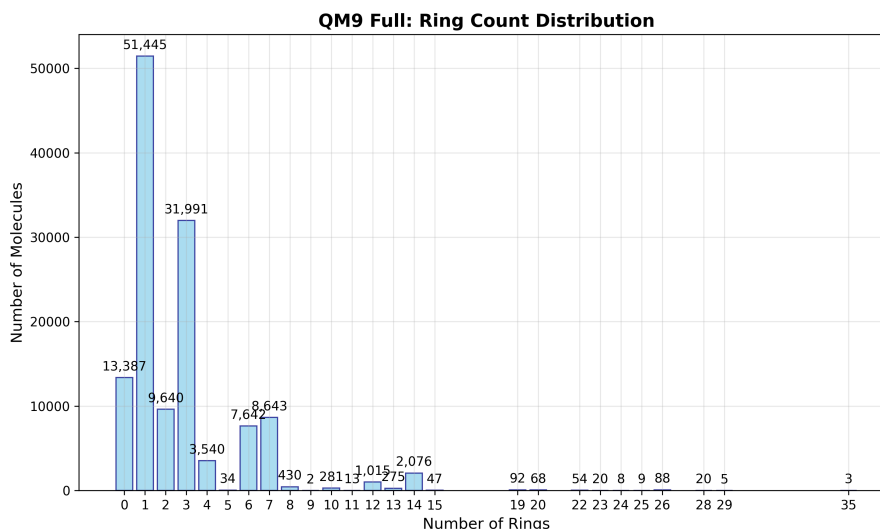


Figure 4.1: Ring count distribution for QM9 full dataset showing the percentage of molecules with different numbers of rings.

Each entry in Table 4.2 is the percentage of molecules with exactly k simple cycles. The most common ring count is $k = 1$ in every split, followed by $k = 3$. There is a long but very sparse tail of higher cycle counts: a small fraction of molecules have $k \geq 10$ cycles (e.g. around 3.1% in the full set), due to highly fused ring systems where multiple small cycles are counted individually. A more detailed version of this table, showing all counts up to $k = 35$, is available in Figure 4.1; however, we truncate it here for clarity, as the higher ring counts are both rare and less informative. The train/validation/test splits closely mirror the full distribution (differences of approximately 1% or less), indicating that the dataset partitions are representative.

Set (size)	Maximum ring length per molecule L (values in %)							
	0 (no rings)	3	4	5	6	7	8	9
Full (130,828)	10.2325	10.5910	11.0733	26.9338	20.4574	11.5258	6.9205	2.2656
Train (97,732)	10.2924	10.4633	11.0015	26.9635	20.5562	11.4824	6.9261	2.3145
Val (20,042)	10.3233	11.0568	11.3013	26.4295	20.3223	11.5458	6.7907	2.2303
Test (13,054)	9.6446	10.8319	11.2609	27.4858	19.9249	11.8201	7.0783	1.9534

Table 4.3: Distribution of molecules by maximum ring length L .

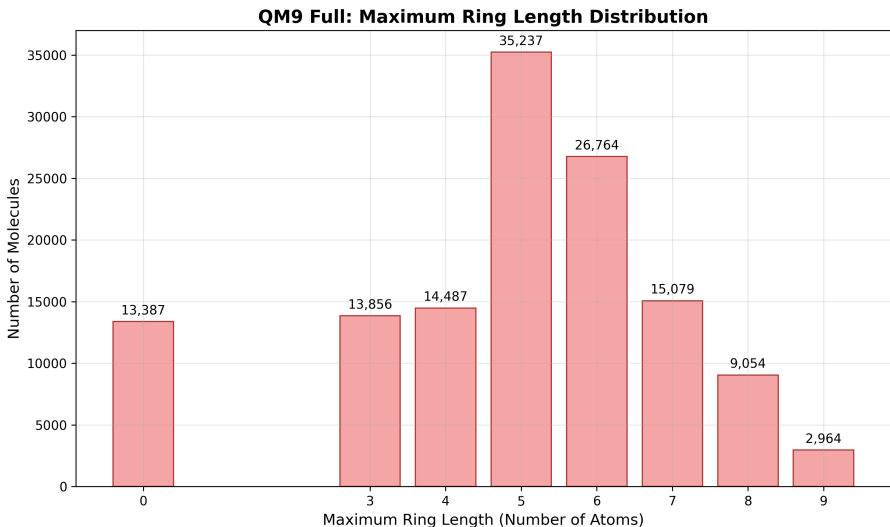


Figure 4.2: Ring length distribution for QM9 full dataset showing the percentage of molecules with different maximum ring lengths.

Table 4.3 shows the distribution of the length L of the largest ring in each molecule. Five-membered ($L = 5$) and six-membered ($L = 6$) rings dominate across all splits. No molecule in QM9 has a ring larger than 9 atoms (the > 9 column is 0% for every split), so any constraint beyond $L = 9$ is non-informative on this dataset. (The acyclic percentage here matches the $k = 0$ percentage above, as expected. So, $L = 0$ is equivalent to acyclic constraint (no rings allowed).)

We also report cumulative distribution functions (CDFs) for *ring count per molecule* and *maximum ring length per molecule*.

Set (size)	Cumulative distribution: ring count $\leq k$ (values in %)										
	≤ 0	≤ 1	≤ 2	≤ 3	≤ 4	≤ 5	≤ 6	≤ 7	≤ 8	≤ 9	≤ 35
Full (130,828)	10.2325	49.5551	56.9236	81.3763	84.0822	84.1081	89.9494	96.5558	96.8845	96.8860	100.0
Train (97,732)	10.2924	49.5058	56.8023	81.3060	84.0155	84.0431	89.8969	96.5283	96.8577	96.8598	100.0
Val (20,042)	10.3233	49.6358	57.1250	81.5937	84.2980	84.3179	90.1008	96.5971	96.9314	96.9314	100.0
Test (13,054)	9.6446	49.8008	57.5226	81.5689	84.2500	84.2730	90.1103	96.6983	97.0124	97.0124	100.0

Table 4.4: Cumulative distribution function (CDF) of ring count per molecule in QM9.

Table 4.4 reports the CDF of ring counts. For example, in the full set, a limit of $k \leq 1$ covers approximately 49.6 % of molecules, and $k \leq 3$ covers approximately 81.4%. The last column (≤ 35) reaches 100% for each split, consistent with the observed maximum ring count (35 rings in full/train; 29 rings in test). See Figures 4.1 and 4.2 for the corresponding histograms.

Set (size)	Cumulative distribution: max ring length $\leq L$ (in %)							
	≤ 0	≤ 3	≤ 4	≤ 5	≤ 6	≤ 7	≤ 8	≤ 9
Full (130,828)	10.2325	20.8235	31.8968	58.8307	79.2881	90.8139	97.7344	100.0
Train (97,732)	10.2924	20.7557	31.7573	58.7208	79.2770	90.7594	97.6855	100.0
Val (20,042)	10.3233	21.3801	32.6814	59.1109	79.4332	90.9789	97.7697	100.0
Test (13,054)	9.6446	20.4765	31.7374	59.2232	79.1482	90.9683	98.0466	100.0

Table 4.5: Cumulative distribution function (CDF) of maximum ring length per molecule in QM9.

Table 4.5 reports the fraction of molecules whose maximum ring length is $\leq L$. For instance, about 79.29% of the full set has no rings larger than 6 atoms ($L \leq 6$), and every QM9 molecule satisfies $L \leq 9$. The training, validation, and test splits closely follow the full distribution.

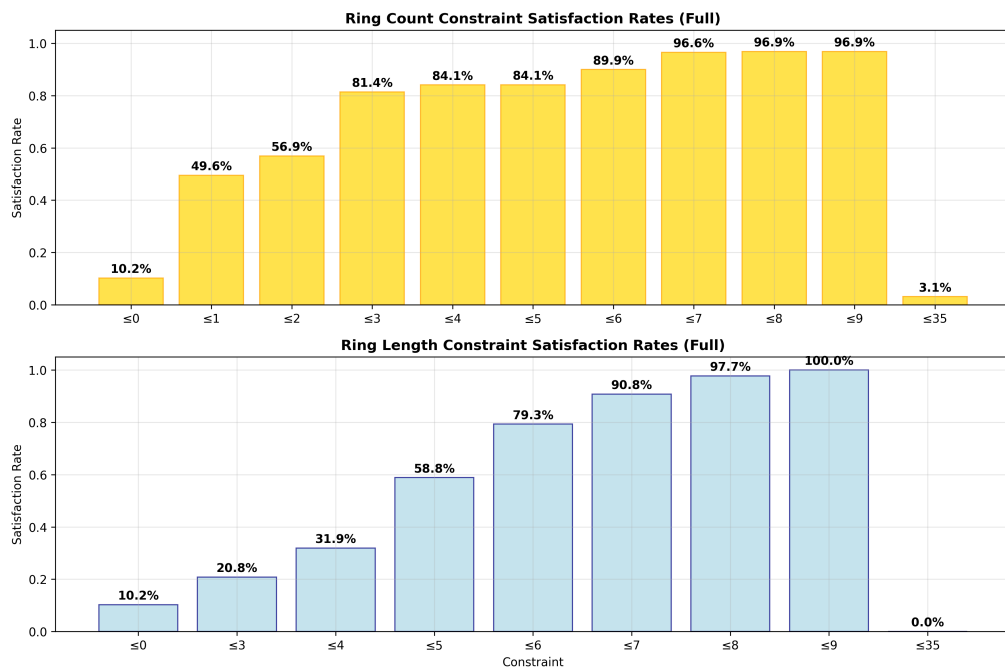


Figure 4.3: Cumulative distributions for the QM9 full dataset. Top: percentage of molecules with ring count $\leq k$. Bottom: percentage of molecules with maximum ring length $\leq L$.

Overall, the dataset’s ring statistics establish the maxima of interest: under our cycle definition, the largest number of rings in any QM9 molecule is 35, and the longest ring size is 9 atoms. These values serve as the upper limits of meaningful constraint settings. Notably, ring count distribution in QM9 peaks at $k = 1$ (39.32%) with a secondary peak at $k = 3$ (24.45%), covering over 80% cumulatively by $k \leq 3$. Only around 3.1% of molecules have 10 or more cycles.

For ring lengths, five- and six-membered rings dominate (26.93% and 20.46% respectively in the full set), and 100% of molecules satisfy $L \leq 9$, so there is a meaningful tail for $L = 7$ or 8 (together around 18% of molecules) but nothing beyond $L = 9$.

Set (size)	Planar (%)
Full (130,828)	100.0
Train (97,732)	100.0
Val (20,042)	100.0
Test (13,054)	100.0

Table 4.6: Planarity satisfaction in the QM9 dataset under the simplified undirected, single-bond graph representation.

In addition, all graphs in QM9 are planar in this representation; hence, planarity is already satisfied at the dataset level, as shown in Table 4.6.

4.4 BASELINE: UNCONSTRAINED SAMPLING

As a reference, we first consider an unconstrained diffusion model (no symbolic constraints during training or sampling). Table 4.7 summarizes the core metrics for this baseline.

Summary metrics	
FCD (lower is better)	1.03 ± 0.01
Unique (%)	97.41 ± 0.15
Novel (%)	88.72 ± 0.09
Valid (%)	98.45 ± 0.09
Disconnected (%)	2.14 ± 0.12
V.U.N. (valid $\times$ unique $\times$ novel) (%)	85.08 ± 0.18

Table 4.7: Baseline metrics for unconstrained training and sampling.

Under this no-constraint baseline, the model achieves high validity and diversity and a low FCD, as shown in Table 4.7. These metrics are consistent with state-of-the-art diffusion models on QM9 (e.g., *DiGress*) [22]. The small fraction of disconnected graphs (around 2.14%) reflects the model’s occasional generation of multiple molecular fragments, which is expected in unconstrained generation without connectivity enforcement.

The unconstrained model outputs 98.45% valid molecules on average, of which 97.41% are unique and 88.72% are novel (not seen in the training set). The Fréchet ChemNet Distance is low (FCD ≈ 1.03), indicating that the generated distribution closely matches the QM9 training distribution.

Beyond the core metrics, structural properties were analyzed to evaluate planarity and ring statistics. As shown in Table 4.8, the unconstrained baseline achieves $99.800 \pm 0.030\%$ planarity across 5 runs—i.e., nearly all generated graphs are planar even without an explicit planarity projector, consistent with QM9 being planar in this representation.

Setting (N)	Planar (%)
No-Constraint (50,000)	99.8000 ± 0.0300

Table 4.8: Planarity satisfaction under the unconstrained baseline (no projector), averaged over 5 runs (mean $\pm$ std).

Table 4.9 and Table 4.10 show the ring count and ring length distributions of 50,000 unconstrained generated molecules, to be compared with the dataset distributions in Section 4.3.

Setting (N)	Ring count per molecule k (in %)										
	0	1	2	3	4	5	6	7	8	9	≥ 10
No-Constraint (50,000)	10.9960	39.9660	7.2340	25.0100	2.0720	0.0120	6.2220	5.1300	0.1380	3.2200	0.0000

Table 4.9: Distribution of generated molecules by ring count k .

The baseline generator’s ring count distribution (Table 4.9) is very close to the QM9 distribution for $k = 0, 1, 2, 3$ rings, and places only a small mass in the high-ring tail. Notably, the model did not produce any molecules with $k \geq 10$ rings (0.0% in the $k \geq 10$ column), in line with the rarity of such cases in the training data. Ring counts of $k = 4$ and above are slightly underrepresented in the generated distribution (16.79% vs. 18.63% in QM9), with a corresponding small shift of mass toward the $k = 3$ category. This suggests the baseline has learned the general distribution well, but with a somewhat softer cutoff on extreme ring counts (e.g., a small fraction (3.22%) of generations have $k = 9$ cycles, whereas in QM9 such extreme cycle counts were virtually nonexistent (only 0.0015% of molecules have $k = 9$ rings, compared to the model’s 3.22%))

	Maximum ring length per molecule L (in %)								
Setting (N)	0	3	4	5	6	7	8	9	≥ 10
No-Constraint (50,000)	10.9960	10.7680	10.7240	26.5900	19.0220	11.7220	7.9560	2.2220	0.0000

Table 4.10: Distribution of generated molecules by maximum ring length L .

Similarly, the baseline’s longest-ring distribution matches QM9 (Table 4.10): it is dominated by 5-atom and 6-atom rings, with smaller contributions from 3, 4, 7, and 8 (none beyond 9).

In fact, the baseline explores the longer-ring tail slightly, producing some cases with $L = 7$ and $L = 8$ rings (together $\approx 19.7\%$), which is consistent with the dataset’s 7+ tail. Every generated molecule respects $L \leq 9$, exactly as in QM9.

To complement these, cumulative satisfaction values were computed. The baseline’s cumulative satisfaction of ring-count and ring-length constraints are given in Table 4.11 and Table 4.12. These cumulative values can be read directly as acceptance rates if a ring-count cap were imposed at sampling time. The trend mirrors the dataset cumulative distribution analyzed in Section 4.3, where we showed that 81.38 % of QM9 molecules have ≤ 3 rings.

Setting (N)	Cumulative: ring count $\leq k$ (in %)										
	≤ 0	≤ 1	≤ 2	≤ 3	≤ 4	≤ 5	≤ 6	≤ 7	≤ 8	≤ 9	≤ 35
No-Constraint (50,000)	10.9960	50.9620	58.1960	83.2060	85.2780	85.2900	91.5120	96.6420	96.7800	100.0	100.0

Table 4.11: Cumulative satisfaction of ring count constraints (per molecule).

Setting (N)	Cumulative: max ring length $\leq L$ (in %)							
	≤ 0	≤ 3	≤ 4	≤ 5	≤ 6	≤ 7	≤ 8	≤ 9
No-Constraint (50,000)	10.9960	21.7640	32.4880	59.0780	78.1000	89.8220	97.7780	100.0

Table 4.12: Cumulative satisfaction of maximum ring length constraints (per molecule).

Table 4.11 shows that about 83.21% of baseline generations have at most 3 rings (compared to 81.38% in QM9 analysis in Table 4.2), and 100% have ≤ 9 rings (since the baseline produced no out-of-distribution cycle counts). Likewise, Table 4.12 indicates that roughly 78.10% of baseline molecules satisfy $L \leq 6$, and all satisfy $L \leq 9$ (exactly matching the dataset’s limits).

In summary, the unconstrained model closely reproduces the training distribution’s ring statistics, with near-perfect fidelity apart from the expected stochastic variations.

4.5 CONSTRAINED SAMPLING WITH PROJECTORS

To isolate the effect of symbolic constraints at generation time, we reuse the same trained model and diffusion hyperparameters as the baseline and enable only sampling-time projectors. Unless specified otherwise, a single projector is activated per experiment (Planarity, Ring Count $\leq K$, or Max Ring Length $\leq L$), and candidate edges that would violate the active constraint are hard-rejected during the reverse process. Results aggregate 5 seeds (10,000 samples/seed).

4.5.1 PLANARITY CONSTRAINT

First, we enable the planarity projector during sampling as a sanity check. Planarity here is graph-theoretic: at each reverse-diffusion step, any candidate edge whose addition would render the graph non-planar is rejected. Although QM9 is planar under our encoding (implicit hydrogens, no multi-edges), the unconstrained sampler occasionally produces non-planar graphs (baseline planarity $99.800 \pm 0.030\%$ over five runs; see Section 4.4, Table 4.8.). With the projector, planarity is driven to $100.000 \pm 0.000\%$ while the core quality metrics remain essentially unchanged relative to the baseline, it shown in Table 4.13 and Table 4.14.

Summary metrics (values in % unless noted)	
FCD (lower is better)	1.03 ± 0.01
Unique (%)	97.40 ± 0.15
Novel (%)	88.72 ± 0.09
Valid (%)	98.45 ± 0.09
Disconnected (%)	2.14 ± 0.13
V.U.N. (%)	85.08 ± 0.19

Table 4.13: Core metrics under the planarity constraint (means $\pm$ std).

Enforcing planarity leaves FCD, validity, uniqueness, and novelty virtually unchanged. This is consistent with QM9’s planar encoding and the baseline’s already high planarity rate: the projector filters the rare non-planar cases and has no effect on the remaining majority of samples. Note that crossings in a plotted layout do not imply non-planarity; we evaluate planarity via graph algorithms [88]. These observations align with prior reports for projector-based enforcement in discrete diffusion [2].

Constraint	Satisfied (%)
Planarity	100.0 ± 0.0
Unconstrained	99.8000 ± 0.0300

Table 4.14: Planarity satisfaction under the constrained model, shown alongside the unconstrained baseline for context.

By construction, the planarity projector achieves 100% satisfaction: every generated graph is planar. Because planarity is intrinsic to QM9, enforcing it at sampling time has essentially no effect on the model’s output distribution. The same near-inert behavior appears for ring-length constraints beyond QM9’s observed range.

4.5.2 RING COUNT CONSTRAINT ($RING\ COUNT \leq K$)

We now apply a constraint on the total ring count k per molecule, using our sampling-time projector to enforce $k \leq K$. We experiment with a range of values K from 0 up to 9. (Recall from the dataset analysis that $K = 9$ already covers more than 96% of QM9, and the maximum k in QM9 is 35.) Table 4.15 reports the core performance metrics as K varies.

K	FCD	Unique (%)	Novel (%)	Valid (%)	V.U.N. (%)	Disconnected (%)
0	10.66 $\pm$ 0.02	81.86 $\pm$ 0.31	90.68 $\pm$ 0.23	99.63 $\pm$ 0.03	73.96 $\pm$ 0.41	1.37 $\pm$ 0.13
1	1.81 $\pm$ 0.01	93.19 $\pm$ 0.31	87.29 $\pm$ 0.13	99.27 $\pm$ 0.14	80.76 $\pm$ 0.41	1.83 $\pm$ 0.16
2	1.41 $\pm$ 0.01	94.51 $\pm$ 0.37	87.22 $\pm$ 0.17	99.17 $\pm$ 0.14	81.75 $\pm$ 0.50	1.90 $\pm$ 0.15
3	1.08 $\pm$ 0.01	96.54 $\pm$ 0.17	88.55 $\pm$ 0.10	98.97 $\pm$ 0.11	84.61 $\pm$ 0.18	2.10 $\pm$ 0.09
4	1.06 $\pm$ 0.01	96.73 $\pm$ 0.17	88.63 $\pm$ 0.08	98.93 $\pm$ 0.12	84.81 $\pm$ 0.20	2.11 $\pm$ 0.11
5	1.06 $\pm$ 0.01	96.73 $\pm$ 0.17	88.63 $\pm$ 0.08	98.93 $\pm$ 0.12	84.81 $\pm$ 0.20	2.11 $\pm$ 0.11
6	1.04 $\pm$ 0.01	97.11 $\pm$ 0.18	88.71 $\pm$ 0.08	98.70 $\pm$ 0.12	85.03 $\pm$ 0.24	2.13 $\pm$ 0.13
7	1.03 $\pm$ 0.01	97.33 $\pm$ 0.14	88.74 $\pm$ 0.09	98.67 $\pm$ 0.11	85.22 $\pm$ 0.19	2.14 $\pm$ 0.13
8	1.03 $\pm$ 0.01	97.33 $\pm$ 0.14	88.74 $\pm$ 0.09	98.67 $\pm$ 0.11	85.23 $\pm$ 0.19	2.14 $\pm$ 0.13
9	1.03 $\pm$ 0.01	97.33 $\pm$ 0.14	88.74 $\pm$ 0.09	98.67 $\pm$ 0.11	85.23 $\pm$ 0.19	2.14 $\pm$ 0.13
Unconstrained	1.03 $\pm$ 0.01	97.41 $\pm$ 0.15	88.72 $\pm$ 0.09	98.45 $\pm$ 0.09	85.08 $\pm$ 0.18	2.14 $\pm$ 0.12

Table 4.15: Core metrics under ring count $\leq K$ (means $\pm$ std) and the unconstrained baseline shown at the bottom.

Imposing a tight cap on ring count significantly shifts the output distribution away from the QM9 baseline. For example, at $K = 0$ (all molecules forced acyclic) the FCD jumps to about 10.66, indicating a large distributional divergence. In that extreme case the model still achieves 99.63% validity, but uniqueness drops to 81.86% as the model is confined to a smaller chemical space. As K increases, the constraints become more permissive and the statistics gradually recover toward the unconstrained baseline. By $K = 3$, FCD has dropped to 1.08, very close to the baseline 1.03; by $K = 7$ or above, FCD $\approx$ 1.03, and all other metrics (validity $\sim$ 98.7%, uniqueness $\sim$ 97.3%, novelty $\sim$ 88.7%) are essentially indistinguishable from the unconstrained case. Throughout the range, validity remains high (all $\geq$ 98.6%) and uniqueness and novelty stay robust. Notably, novelty is actually higher for very strict constraints ($\sim$ 90.68% at $K = 0$) than in the baseline (88.72%), indicating that forcing the model into a narrower domain (acyclic molecules) encourages exploration of structures outside the original training distribution, resulting in higher novelty rates. As K grows, novelty settles back to $\approx$ 88.7%, showing that we are not sacrificing novel outputs when constraints are moderate.

Distribution of molecules with their exact ring counts under total ring count K (values in %)											
Ring Count $\leq K$	0-ring	1-ring	2-rings	3-rings	4-rings	5-rings	6-rings	7-rings	8-rings	9-rings	≥ 10
Ring Count ≤ 0	100.0	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Ring Count ≤ 1	10.9960	89.0040	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Ring Count ≤ 2	10.9960	72.2520	16.7520	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Ring Count ≤ 3	10.9960	39.9660	10.4960	38.5420	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Ring Count ≤ 4	10.9960	39.9660	8.4580	37.1960	3.3840	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Ring Count ≤ 5	10.9960	39.9660	8.4580	37.1880	3.3720	0.0200	0.0000	0.0000	0.0000	0.0000	0.0000
Ring Count ≤ 6	10.9960	39.9660	7.2340	29.8120	2.7140	0.0140	9.2640	0.0000	0.0000	0.0000	0.0000
Ring Count ≤ 7	10.9960	39.9660	7.2340	25.0120	2.3520	0.0120	7.9080	6.5200	0.0000	0.0000	0.0000
Ring Count ≤ 8	10.9960	39.9660	7.2340	25.0120	2.2240	0.0120	7.9060	6.4700	0.1800	0.0000	0.0000
Ring Count ≤ 9	10.9960	39.9660	7.2340	25.0120	2.2240	0.0120	7.9040	6.4620	0.1800	0.0100	0.0000
QM9 Dataset	10.2325	39.3226	7.3685	24.4527	2.7058	0.0260	5.8413	6.6064	0.3287	0.0015	3.1140
Unconstrained	10.9960	39.9660	7.2340	25.0100	2.0720	0.0120	6.2220	5.1300	0.1380	3.2200	0.0000

Table 4.16: Exact ring count distributions under limit K , with QM9 and the unconstrained baseline shown at the bottom for direct comparison.

Each row of Table 4.16 gives the percentage of outputs with an exact ring count, when using a particular K constraint. By design, no molecule exceeds the cap (e.g. for $K = 3$, there are 0% with 4 or more rings). As K increases, the distribution allocates more weight to higher ring counts that become allowed, gradually approaching the unconstrained profile while keeping any forbidden tail probability at 0%. For example, at $K = 7$ the permitted counts (0–7) collectively account for almost 100% of the molecules, with a distribution nearly identical to the baseline’s for $k \leq 7$. By $K = 9$, the distribution of ring counts is essentially the same as the unconstrained case (compare the $K = 9$ row with the “Unconstrained” row in Table 4.16, except that $K = 9$ strictly prevents any output with more than 9 rings (which, as noted, the baseline didn’t produce anyway). In summary, looser ring-count constraints (larger K) leave the model’s output distribution almost unchanged, whereas strict constraints (smaller K) remove the high-ring-count tail entirely and reallocate that probability mass into the lower- k bins.

K	Property satisfied (%)
0, 1, 2, 3, 4, 5, 6, 7, 8, 9	100.0 $\pm$ 0.0

Table 4.17: Constraint satisfaction results for ring count $\leq K$ under the constrained generation setting.

The ring-count projector achieves perfect satisfaction for every tested K . In other words, 100% of the generated molecules respect the specified cap on k . This confirms that our constraint projector is effectively enforcing the limit during sampling without any exceptions.

To push the ring-count projector beyond the dataset’s natural limits, we conducted a non-binding test with a very high threshold $K = 40$. (No QM9 molecule has anywhere near 40 rings (the maximum observed is 35) so $K = 40$ imposes a constraint well above the dataset range.) Intuitively, this setting should behave almost identically to the unconstrained baseline, since the model will virtually never attempt to produce a structure with over 40 cycles. Tables 4.18 and 4.19 confirm this: the core metrics are effectively the same as the no-constraint case, and the ring-count distribution is unchanged.

Ring Count $\leq K$	FCD	Unique (%)	Novel (%)	Valid (%)	V.U.N. (%)	Disconnected (%)
Ring Count ≤ 40 (50,000)	1.03 ± 0.01	97.41 ± 0.15	88.72 ± 0.09	98.45 ± 0.09	85.08 ± 0.18	2.14 ± 0.12
Unconstrained	1.03 ± 0.01	97.41 ± 0.15	88.72 ± 0.09	98.45 ± 0.09	85.08 ± 0.18	2.14 ± 0.12

Table 4.18: Core metrics under ring count ≤ 40 (non-binding high threshold).

Ring Count $\leq K$	0	1	2	3	4	5	6	7	8	9	≥ 10
Ring Count ≤ 40 (50,000)	10.9960	39.9660	7.2340	25.0100	2.0720	0.0120	6.2220	5.1300	0.1380	3.2200	0.0000
QM9 Dataset	10.2325	39.3226	7.3685	24.4527	2.7058	0.0260	5.8413	6.6064	0.3287	0.0015	3.1140
Unconstrained	10.9960	39.9660	7.2340	25.0100	2.0720	0.0120	6.2220	5.1300	0.1380	3.2200	0.0000

Table 4.19: Distribution of generated molecules by ring count k under ring count ≤ 40 .

With $K = 40$, the constraint never actually comes into play; none of the 50,000 samples approached the 40-ring limit (the highest ring count observed was 9, identical to the baseline). The summary metrics (FCD ≈ 1.03 , Validity $\approx 98.45\%$, etc.) match the unconstrained baseline within random variation. The ring count distribution (Table 4.19) is essentially identical to the baseline’s: for instance, $k = 3$ is about 25.01% and the model still generates no molecules with $k \geq 10$. This sanity-check experiment validates that the projector does not introduce any drift or artifacts when the specified cap lies above the model’s natural generative range – in other words, an overly loose ring-count constraint has no effect (aside from trivially being satisfied 100% of the time), confirming the projector’s robustness.

A selection of generated molecules illustrating these effects can be found in Fig. 4.4. The figure visually summarizes how the imposed ring-count constraint shapes molecular topology across different thresholds $K = 1, 2, 3, 4$. As K increases, the model transitions from simple monocyclic or acyclic motifs to more complex polycyclic and bridged structures, demonstrating that the constraint mechanism effectively modulates structural diversity without sacrificing chemical validity.

Generated Molecules under Ring Constraints ($K = 1, 2, 3, 4$)

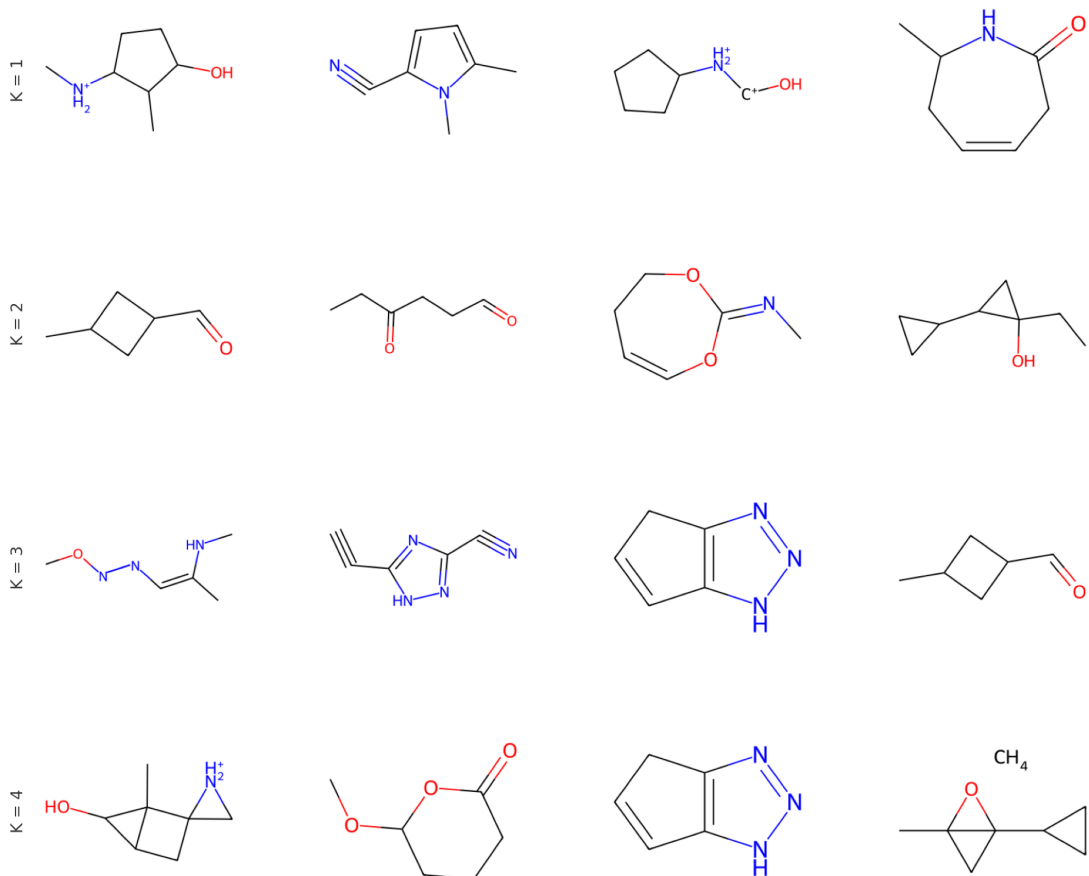


Figure 4.4: Representative molecules generated under different ring-count limits $K = 1, 2, 3, 4$. Each row corresponds to a fixed constraint value K , showing four randomly sampled valid outputs from the proposed diffusion model. As K increases, the model produces more polycyclic and structurally complex molecules.

4.5.3 RING LENGTH CONSTRAINT ($\text{MAXIMUM RING LENGTH} \leq L$)

We next apply a constraint on the maximum ring length $L_{\max}$ in each molecule, forcing all rings to be of size L or smaller during generation. We consider L values from 0 (which allows no rings at all, effectively forcing acyclic molecules) up to 9 (the largest ring size present in QM9, covering 100% of the dataset). Table 4.20 summarizes the performance results across varying values of L .

	L	FCD	Unique (%)	Novel (%)	Valid (%)	V.U.N. (%)	Disconnected (%)
	0	10.66 $\pm$ 0.02	81.86 $\pm$ 0.31	90.68 $\pm$ 0.23	99.63 $\pm$ 0.03	73.96 $\pm$ 0.41	1.37 $\pm$ 0.13
	3	5.18 $\pm$ 0.03	90.76 $\pm$ 0.44	92.35 $\pm$ 0.30	99.45 $\pm$ 0.06	83.36 $\pm$ 0.53	1.57 $\pm$ 0.13
	4	3.95 $\pm$ 0.01	94.39 $\pm$ 0.28	92.95 $\pm$ 0.30	98.88 $\pm$ 0.06	86.76 $\pm$ 0.49	1.72 $\pm$ 0.10
	5	1.37 $\pm$ 0.02	95.80 $\pm$ 0.25	88.86 $\pm$ 0.23	98.67 $\pm$ 0.07	83.99 $\pm$ 0.41	1.98 $\pm$ 0.11
	6	1.08 $\pm$ 0.01	96.77 $\pm$ 0.16	88.58 $\pm$ 0.09	98.55 $\pm$ 0.07	84.47 $\pm$ 0.21	2.13 $\pm$ 0.14
	7	1.05 $\pm$ 0.01	97.23 $\pm$ 0.16	88.90 $\pm$ 0.06	98.48 $\pm$ 0.08	85.12 $\pm$ 0.18	2.15 $\pm$ 0.13
	8	1.02 $\pm$ 0.01	97.39 $\pm$ 0.14	88.76 $\pm$ 0.09	98.45 $\pm$ 0.08	85.10 $\pm$ 0.18	2.14 $\pm$ 0.12
	9	1.03 $\pm$ 0.01	97.41 $\pm$ 0.15	88.72 $\pm$ 0.09	98.45 $\pm$ 0.09	85.08 $\pm$ 0.18	2.14 $\pm$ 0.12
	Unconstrained	1.03 $\pm$ 0.01	97.41 $\pm$ 0.15	88.72 $\pm$ 0.09	98.45 $\pm$ 0.09	85.08 $\pm$ 0.18	2.14 $\pm$ 0.12

Table 4.20: Core metrics under max ring length $\leq L$ (means $\pm$ std) with the unconstrained baseline for comparison.

Small L values (strict ring-length caps) cause noticeable distribution shifts. For example, at $L = 3$ (all rings must be triangles or acyclic), FCD rises to 5.18 ± 0.03 , significantly higher than the baseline. Even $L = 4$ (allowing rings of size 3 or 4, plus acyclic structures) yields $\text{FCD} \approx 3.95$. These high FCDs indicate the model is being pushed into parts of chemical space underrepresented in QM9 (e.g., molecules composed of only very small rings). In those cases, validity remains around 99% and novelty actually increases: for $L = 3$ or 4, novelty is around 93%, compared to 88.72% baseline, indicating the model explores novel chemical space when constrained to small rings. As L relaxes, the model’s outputs realign with the training distribution. By $L = 6$, FCD is down to approximately 1.08 and by $L = 7$ it’s around 1.05; at $L = 8$ and $L = 9$, $\text{FCD} \approx 1.02$ matching the baseline. In fact, for $L \geq 6$ all other metrics (validity 98.5%, uniqueness 97.3%, novelty 88.7%) are within normal variation of the unconstrained case. In summary, moderate ring-length constraints ($L = 5$ or 6) achieve near-baseline fidelity, whereas very strict constraints ($L = 3$ or 4) push the model to explore novel but distribution-shifted outputs (reflected in higher FCD and novelty).

Max Ring Length $\leq L$	Distribution of molecules with their exact ring lengths under maximum ring length L_{max} (values in %)								
	0 (no rings)	3-atom rings	4-atom rings	5-atom rings	6-atom rings	7-atom rings	8-atom rings	9-atom rings	≥ 10
Max Ring Length ≤ 0	100.0	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Max Ring Length ≤ 3	54.3420	45.6580	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Max Ring Length ≤ 4	38.5180	29.2780	32.2040	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Max Ring Length ≤ 5	17.0060	16.8180	18.5360	47.6400	0.0000	0.0000	0.0000	0.0000	0.0000
Max Ring Length ≤ 6	12.1240	12.3840	13.3540	35.0960	27.0420	0.0000	0.0000	0.0000	0.0000
Max Ring Length ≤ 7	11.2640	11.1780	11.3160	30.4360	21.2640	14.5420	0.0000	0.0000	0.0000
Max Ring Length ≤ 8	11.0220	10.8240	10.7960	27.1160	19.4740	12.1160	8.6520	0.0000	0.0000
Max Ring Length ≤ 9	10.9960	10.7680	10.7240	26.5900	19.0220	11.7220	7.9560	2.2220	0.0000
QM9 Dataset	10.2325	10.5910	11.0733	26.9338	20.4574	11.5258	6.9205	2.2656	0.0000
Unconstrained	10.9960	10.7680	10.7240	26.5900	19.0220	11.7220	7.9560	2.2220	0.0000

Table 4.21: Exact maximum ring length distributions under limit L , with QM9 and the unconstrained baseline.

Each row of Table 4.21 shows the conditional distribution of the largest ring lengths in the outputs when enforcing a given L , also with QM9 and the unconstrained baseline shown at the bottom for direct comparison. For example, under $L = 6$ (row “Max 6-atom rings”), the distribution of $L_{\max}$ in generated molecules concentrates at 5 and 6: about 35.10% of molecules end up with a 5-membered ring as their largest, and 27.04% with a 6-membered ring, while no 7+ rings occur. As L increases, the distribution “fills out” to recover the unconstrained shape. By the time we reach $L = 9$, the profile of $L_{\max}$ (last row) is identical to the baseline’s, since allowing up to 9-membered rings imposes essentially no restriction for QM9. In other words, loose length caps ($L = 8, 9$) yield an output distribution nearly the same as unconstrained, whereas tight caps skew the distribution toward the allowed smaller rings (e.g. $L = 3$ yields only triangles, which constituted a minority in QM9).

L	Property satisfied (%)
0, 3, 4, 5, 6, 7, 8, 9	100.0 $\pm$ 0.0

Table 4.22: Constraint satisfaction results for enforcing a maximum ring length $\leq L$ during constrained generation.

The ring-length projector, like the ring-count one, enforces the constraint perfectly in all tested settings (see Table 4.22). No generated molecule contains a ring longer than the imposed L ; the projector simply **rejects** any candidate edge whose insertion would create a cycle exceeding length L . For instance, under $L = 5$, if the model proposes an edge that would close a 6-cycle, that edge is deterministically rejected and added to the global blocklist, ensuring that the final output always satisfies $L_{\max} \leq 5$. The 100% satisfaction rates observed in our experiments confirm the effectiveness of this projector mechanism.

Analogous to the ring-count sanity test, we also experiment with a non-binding ring-length constraint beyond the dataset’s domain: $L = 12$. Since no QM9 molecule has a ring larger than 9 atoms, a constraint of $L_{\max} \leq 12$ should be effectively inactive. We expect the model’s behavior to remain unchanged in this scenario. Tables 4.23 and 4.24 verify this expectation: the results for $L = 12$ are essentially the same as the $L = 9$ case (and baseline), and no rings longer than 9 appear despite the higher allowance.

Max Ring Length $\leq L$	FCD	Unique (%)	Novel (%)	Valid (%)	V.U.N. (%)	Disconnected (%)
Max Ring Length ≤ 12	1.03 $\pm$ 0.01	97.41 $\pm$ 0.15	88.72 $\pm$ 0.09	98.45 $\pm$ 0.09	85.08 $\pm$ 0.18	2.14 $\pm$ 0.12
Unconstrained	1.03 $\pm$ 0.01	97.41 $\pm$ 0.15	88.72 $\pm$ 0.09	98.45 $\pm$ 0.09	85.08 $\pm$ 0.18	2.14 $\pm$ 0.12

Table 4.23: Core metrics under max ring length ≤ 12 (non-binding high threshold).

With $L_{\max} = 12$, the model’s statistics are indistinguishable from the unconstrained baseline: FCD ≈ 1.03 , validity $\approx 98.45\%$, etc., matching those in Table 4.7. The distribution of largest ring lengths (Table 4.24) is exactly the same as for $L_{\max} \leq 9$ (Table 4.10): the outputs contain five- and six-membered rings most frequently, a small fraction of 7- and 8- membered rings, and none above 9. This confirms that allowing up to 12-member rings does not lead the model to generate any 10, 11, or 12 atom cycles. In effect, $L = 12$ behaves as if no constraint were applied at all, and the projector simply reports 100% satisfaction without needing to intervene. This provides further evidence that our diffusion model does not spontaneously stray beyond the training distribution’s support, and that the projector introduces no side effects when the constraint is trivially satisfied by the model’s natural outputs.

Max Ring Length $\leq L$	0	3	4	5	6	7	8	9	≥ 10
Max Ring Length ≤ 12	10.996	10.768	10.724	26.590	19.022	11.722	7.956	2.222	0.000
QM9 Dataset	10.2325	10.5910	11.0733	26.9338	20.4574	11.5258	6.9205	2.2656	0.0000
Unconstrained	10.9960	10.7680	10.7240	26.5900	19.0220	11.7220	7.9560	2.2220	0.0000

Table 4.24: Distribution of generated molecules by maximum ring length $L_{\max}$ under max ring length ≤ 12 .

A selection of generated molecules under this constraint is shown in Fig. 4.5. The figure illustrates how a maximum ring-length limit shapes cyclic topology across $L = 3, 4, 5, 8$. At the strictest setting ($L = 3$), the model is limited to very small cycles, producing compact, tightly constrained motifs. As L increases, the generator allows larger rings and more flexible cyclic backbones, yielding broader geometric configurations. This gradual relaxation demonstrates that the projector effectively regulates cycle size while preserving chemical validity and structural diversity.

Generated Molecules under Ring Length Constraints ($L = 3, 4, 5, 8$)

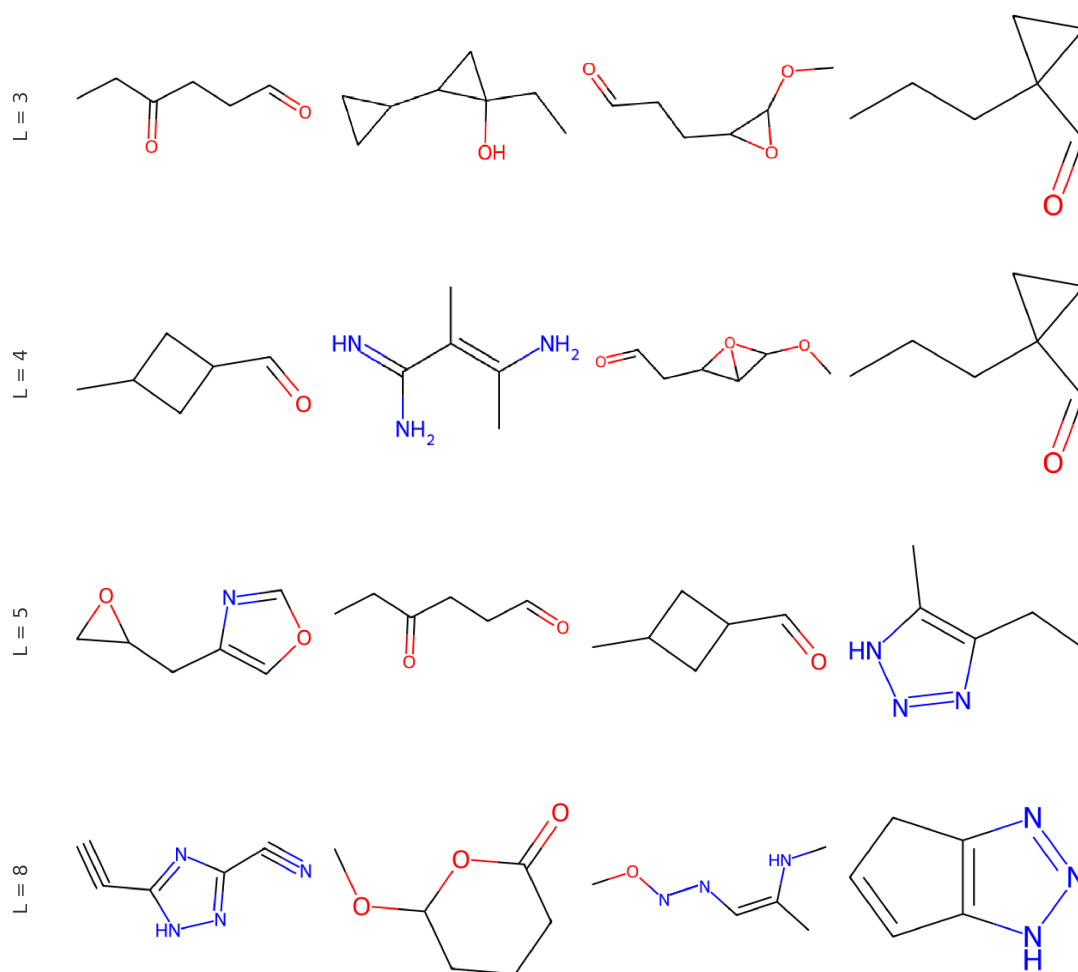


Figure 4.5: Representative molecules generated under different ring-length limits $L = 3, 4, 5, 8$. Each row corresponds to a fixed constraint value L , showing four randomly sampled valid outputs from the proposed diffusion model. As L increases, the model allows progressively larger ring structures, exhibiting greater topological flexibility.

5

Discussion & Takeaways

Our experiments demonstrate that sampling-time projectors can enforce symbolic structural constraints in molecular graph generation with essentially perfect compliance and minimal disruption to the generative model’s performance. In this section, we discuss the implications on constraint satisfaction, distributional shifts, and the balance between fidelity (the generated samples’ quality and resemblance to the training distribution) and novelty, and then summarize key takeaways.

CONSTRAINT SATISFACTION AND CORRECTNESS

Across all tested conditions (planarity, limits on the total number of rings per molecule, and limits on the maximum ring length), the reported satisfaction rate is 100.0% in every case. The projectors integrate into the reverse-diffusion sampling procedure such that any violation of a constraint is immediately corrected, and no invalid samples survive. This behavior also holds for non-binding edge cases: for example, when we set thresholds far beyond the dataset’s range (e.g., “max ring count ≤ 40 ” or “max ring length ≤ 12 atoms”), satisfaction is trivially 100% because the baseline model already stays within those limits.

Importantly, using these projectors does not compromise chemical correctness: validity remains near 98–99% in all experiments, and the model still generates connected, chemically valid molecules. This outcome is expected; by construction, the projector never introduces invalid bonds; it only removes or prevents disallowed ones.

DISTRIBUTIONAL SHIFTS AND CONSTRAINT EFFECTS

While constraint satisfaction is binary (and perfectly achieved), applying constraints does influence the distribution of generated molecules. We observe a clear trade-off between constraint strictness and distributional fidelity, which we illustrate through detailed comparisons of ring statistics across the QM9 dataset, an unconstrained baseline generation, and constrained generations at selected strictness levels.

RING LENGTH DISTRIBUTIONS

Below, Table 5.1 compares the distribution of the *maximum ring length per molecule* in different settings. We report QM9, the unconstrained sampling, and constrained sampling at “maximum ring length ≤ 0 ”, “ ≤ 3 ”, “ ≤ 6 ”, and “ ≤ 9 ”.

The unconstrained (baseline) generation closely follows the dataset: five-membered rings account for 26.59% vs. 26.93% in QM9; six-membered rings 19.02% vs. 20.46%; and the 7+ tail is 21.90% vs. 20.71% in QM9. Acyclic molecules are also similar (10.996% vs. 10.2325%).

Setting	Maximum ring length (atoms) (values in %)								
	0	3	4	5	6	7	8	9	≥ 10
QM9	10.2325	10.5910	11.0733	26.9339	20.4574	11.5258	6.9205	2.2656	0.0000
Unconstrained Generation	10.9960	10.7680	10.7240	26.5900	19.0220	11.7220	7.9560	2.2220	0.0000
Max ring length ≤ 0	100.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Max ring length ≤ 3	54.3420	45.6580	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Max ring length ≤ 6	12.1240	12.3840	13.3540	35.0960	27.0420	0.0000	0.0000	0.0000	0.0000
Max ring length ≤ 9	10.9960	10.7680	10.7240	26.5900	19.0220	11.7220	7.9560	2.2220	0.0000

Table 5.1: Distribution of the maximum ring length per molecule across QM9, unconstrained, and constrained generations.

Setting	Core metrics (mean $\pm$ std) — Max ring length $\leq L$					
	FCD	Valid (%)	Unique (%)	Novel (%)	V.U.N. (%)	Disconnected (%)
Unconstrained Generation	1.03 $\pm$ 0.01	98.45 $\pm$ 0.09	97.41 $\pm$ 0.15	88.72 $\pm$ 0.09	85.08 $\pm$ 0.18	2.14 $\pm$ 0.12
Max ring length ≤ 0	10.66 $\pm$ 0.02	99.63 $\pm$ 0.03	81.86 $\pm$ 0.31	90.68 $\pm$ 0.23	73.96 $\pm$ 0.41	1.37 $\pm$ 0.13
Max ring length ≤ 3	5.18 $\pm$ 0.03	99.45 $\pm$ 0.06	90.76 $\pm$ 0.44	92.35 $\pm$ 0.30	83.36 $\pm$ 0.53	1.57 $\pm$ 0.13
Max ring length ≤ 6	1.08 $\pm$ 0.01	98.55 $\pm$ 0.07	96.77 $\pm$ 0.16	88.58 $\pm$ 0.09	84.47 $\pm$ 0.21	2.13 $\pm$ 0.14
Max ring length ≤ 9	1.03 $\pm$ 0.01	98.45 $\pm$ 0.09	97.41 $\pm$ 0.15	88.72 $\pm$ 0.09	85.08 $\pm$ 0.18	2.14 $\pm$ 0.12

Table 5.2: Core metrics comparison across selected ring-length limits. Baseline included for reference.

Strict limits (e.g., “max ring length ≤ 0 ” or “ ≤ 3 ”) collapse the distribution into very small rings (or no rings at all), strongly diverging from QM9. For instance, at “ ≤ 3 ”, 45.658% of molecules have their largest ring of size 3 and 54.342% are acyclic. As expected, this shift carries a high fidelity cost (FCD = 5.18 ± 0.03 at “ ≤ 3 ” and 10.66 ± 0.02 at “ ≤ 0 ”; Table 5.2).

Moderate limits such as “max ring length ≤ 6 ” remove the 7+ tail entirely and reallocate mass primarily to 5- and 6-membered rings (35.096% and 27.042%, respectively). Distributional drift is contained and core metrics remain close to baseline (FCD = 1.08 ± 0.01 vs. ≈ 1.03 without constraints; Table 5.2).

Loose limits like “max ring length ≤ 9 ” are effectively non-binding in our setting and match the unconstrained profile almost exactly.

RING COUNT DISTRIBUTIONS

Similarly below, Table 5.3 reports the distribution of the *maximum number of rings per molecule* (exact counts 0, . . . , 9) for QM9, the unconstrained sampling, and constrained sampling at “maximum ring count ≤ 0 ”, “ ≤ 1 ”, “ ≤ 6 ”, and “ ≤ 9 ”.

Setting	Max ring count per molecule (values in %)										
	0	1	2	3	4	5	6	7	8	9	≥ 10
QM9	10.2325	39.3226	7.3685	24.4527	2.7059	0.0259	5.8413	6.6064	0.3287	0.0015	3.1140
Unconstrained Generation	10.9960	39.9660	7.2340	25.0100	2.0720	0.0120	6.2220	5.1300	0.1380	3.2200	0.0000
Max ring count ≤ 0	100.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Max ring count ≤ 1	10.9960	89.0040	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Max ring count ≤ 6	10.9960	39.9660	7.2340	29.8120	2.7140	0.0140	9.2640	0.0000	0.0000	0.0000	0.0000
Max ring count ≤ 9	10.9960	39.9660	7.2340	25.0120	2.2240	0.0120	7.9040	6.4620	0.1800	0.0100	0.0000

Table 5.3: Distribution of the maximum number of rings per molecule across QM9, unconstrained, and constrained generations.

Setting	Core metrics (mean $\pm$ std) — Ring count $\leq K$					
	FCD	Valid (%)	Unique (%)	Novel (%)	V.U.N. (%)	Disconnected (%)
Unconstrained Generation	1.03 ± 0.01	98.45 ± 0.09	97.41 ± 0.15	88.72 ± 0.09	85.08 ± 0.18	2.14 ± 0.12
Max ring count ≤ 0	10.66 ± 0.02	99.63 ± 0.03	81.86 ± 0.31	90.68 ± 0.23	73.96 ± 0.41	1.37 ± 0.13
Max ring count ≤ 1	1.81 ± 0.01	99.27 ± 0.14	93.19 ± 0.31	87.29 ± 0.13	80.76 ± 0.41	1.83 ± 0.16
Max ring count ≤ 6	1.04 ± 0.01	98.70 ± 0.12	97.11 ± 0.18	88.71 ± 0.08	85.03 ± 0.24	2.13 ± 0.13
Max ring count ≤ 9	1.03 ± 0.01	98.67 ± 0.11	97.33 ± 0.14	88.74 ± 0.09	85.23 ± 0.19	2.14 ± 0.13

Table 5.4: Core metrics comparison across selected ring-count limits. Baseline included for reference.

Relative to QM9, the unconstrained generator exhibits a slightly heavier high-count tail (max ring count ≥ 4 : 16.7940% vs. 15.5097% in QM9), driven primarily by a spike at 9 rings (3.2200% vs. 0.0015%). It also shows a small shift toward 3 rings (25.0100% vs. 24.4527%). Acyclic and single-ring shares remain close to the dataset (10.9960% and 39.9660% vs. 10.2325% and 39.3226%).

Strict limits such as “max ring count ≤ 0 ” or “ ≤ 1 ” severely alter the landscape. At “ ≤ 1 ”, 89.004% of molecules are single-ring and the rest acyclic; multi-ring motifs vanish. This shift coincides with a large fidelity cost (FCD = 1.81 ± 0.01 at “ ≤ 1 ” and 10.66 ± 0.02 at “ ≤ 0 ”; Table 5.4).

Moderate limits such as “max ring count ≤ 6 ” prune the extreme tail entirely (7+ rings: 0.000% vs. 8.488% in the unconstrained baseline) and redistribute mass primarily to 3 and 6 rings (29.8120% and 9.2640% vs. 25.0100% and 6.2220%). Core metrics remain close to the baseline (FCD = 1.0362 ± 0.0095 vs. ≈ 1.03 without constraints; Table 4.15). Validity and uniqueness stay high at this setting (e.g., Valid $\approx 98.7\%$, Unique $\approx 97.1\%$).

Loose limits like “max ring count ≤ 9 ” are non-binding in aggregate: the total high-count tail mass matches the unconstrained run almost exactly (max ring count ≥ 4 : 16.7920% vs. 16.7940%). Small bin-level deviations appear within the tail (notably the rare 9-ring bin), but the overall shape and the low-count mass remain aligned with the unconstrained distribution.

In summary, the unconstrained model faithfully reproduces the QM9 distribution, exhibiting only minor deviations in the high-ring-count tail. Imposing constraints alters this landscape according to their severity. *Stricter limits* actively distort the distribution, collapsing complex multi-ring structures into simpler motifs and significantly degrading fidelity. Conversely, *more moderate limits* act as surgical interventions, pruning only the extreme outliers while redistributing probability to chemically common motifs, thereby preserving core fidelity metrics comparable to the baseline. Finally, *looser limits* are effectively inert, aligning with the model’s natural generation range and yielding negligible distributional changes.

NOVELTY AND DIVERSITY

Diversity (uniqueness) remains high under all constrained settings we tried. Uniqueness stays above 94% for moderate and loose limits, only dipping to approximately 81–90% in the most extreme cases (no rings allowed or very small maximum ring length) where the chemical space is inherently limited. Novelty actually increases under stricter limits; for example, with “max ring length ≤ 4 ”, novelty was about 92.95% (versus 88.72% baseline). Intuitively, forbidding

common 5–6 membered rings pushes the model toward less frequent motifs (multiple 3–4 membered rings or acyclic structures), yielding a larger fraction of molecules outside the training set. In other words, constraints can steer the model into more novel regions of chemical space. This is reflected in higher FCD, as noted, because those novel molecules deviate from the original distribution. However, even when novelty spiked to $\sim 92\text{--}93\%$, validity remained near 99%. The projector enforces the structural limits while the underlying diffusion model maintains chemical plausibility as it explores new configurations. Practically, if the goal is to generate novel molecules with specific structural properties, a relatively tight limit is a viable strategy — the price is a higher FCD indicating that those molecules are less “like QM9” overall. Meanwhile, uniqueness remains close to 100% in most settings, indicating no collapse.

PLANARITY AS A SANITY CHECK

Applying the planarity constraint served primarily as a sanity check. QM9 molecules are already planar in our representation, so the projector had essentially no violations to reject. As expected, the metrics with and without the planarity projector were indistinguishable (FCD ~ 1.03 , validity $\sim 98.45\%$, etc.). This confirms two points: (i) inserting a sampling-time projector does not degrade performance when all generated samples inherently satisfy the constraint; (ii) the implementation behaves as expected for constraints that never trigger rejections. We observe the same pattern for trivially satisfied high-threshold ring constraints (e.g., “max ring count ≤ 40 ” or “max ring length ≤ 12 ”), where the projector almost never rejects a sample because the dataset never approaches these limits. Taken together, these results show that the projection mechanism is robust: if a constraint is too loose (thresholds far beyond the dataset range), the projector has effectively no influence and introduces no artifacts.

LIMITATIONS & DESIGN CHOICES

Our approach inherits both advantages and limitations from the *ConStruct* framework [2]. A clear limitation is that the projector only addresses the specific structural constraints we encode. For instance, we do *not* enforce any chemical valence rules or atom-specific constraints during generation, hence the model can produce chemically implausible molecules (like an atom with too many or too few bonds) as long as they satisfy the structural criteria. While post-filter validity is high ($\sim 98\%$), about 1–2% of outputs are invalid due to valence errors or disconnected structures (Table 4.7). One could add a valence-check projector, but we chose to isolate the effect of planarity and the two ring-based limits (max ring count, max ring length).

Another design choice is to focus on *hard* constraints (satisfied or not). This can bias topology distribution in non-obvious ways. For example, forbidding all rings makes the model favor trees, which may inflate novelty (many distinct trees not seen in training) but harms distribution resemblance (FCD becomes much worse). In some cases the projector can reduce output diversity because alternatives would violate the constraint which is an inherent trade-off when limiting the generative space. Particularly in QM9, the largest rings have length 9, so “maximum ring length ≤ 9 ” is effectively a no-op; conversely, limits below 5 are very strict because 5–6 membered rings are extremely common (see Table 4.3). Similarly, most QM9 molecules have at most 3 rings; pushing the maximum ring count below 3 imposes a significant change, whereas allowing 4 or more rings scarcely affects anything (see Table 4.2). In other datasets, the same projectors will scale their effect with how prevalent the forbidden structures are in the original distribution.

Lastly, because our method rejects edges that would violate the constraint during sampling, it can leave some atoms under-connected, which contributes to a modest increase in disconnected outputs. Other constrained-diffusion approaches may incorporate the constraints directly into the diffusion dynamics or the training objective, which can preserve global structure more effectively but typically require more complex and constraint-specific formulations. Our choice of sampling-time projectors prioritizes simplicity and generality, while acknowledging that this design does not guarantee preservation of all distributional characteristics.

TAKEAWAYS

Despite these limitations, the results paint a clear picture of what sampling-time projectors contribute to discrete diffusion models. In practice, they offer several tangible advantages:

Projectors are effective. They enforce the specified graph constraints with near-perfect success (100% compliance in our experiments) without retraining. Even non-local constraints such as “maximum ring count ≤ 3 ” are satisfied for all generated samples.

Projectors are modular. The approach is plug-in. We demonstrated planarity, ring count, and ring length, but any property checkable on a graph can be enforced. Constraints are easy to define and adjust, enabling controlled steering.

Projectors provide structural control and insight. Tightening a limit directly removes unwanted structures at the cost of some distributional change; loosening it recovers the original behavior. This gives users explicit control over specificity versus fidelity/diversity and serves as a diagnostic tool for learned model biases.

6

Conclusion and Future Work

This thesis began with a simple question: *can we control molecular structure during generation without retraining the model?* Our experiments on the QM9 dataset [30] demonstrate that the answer is yes—we can achieve perfect constraint compliance (100% satisfaction) while maintaining chemical validity rates of 98–99%. By filtering edge insertions during the reverse diffusion process, we can push generations toward desired ring statistics without modifying the underlying *ConStruct* framework [2, 22].

We discovered a clear spectrum of effects based on constraint strictness. Strict constraints (e.g., maximum ring count ≤ 0 , maximum ring length ≤ 3) dramatically alter distributions, increasing FCD values to 5–10 while boosting novelty to 90–93% as the model explores uncommon structural motifs. Moderate constraints (e.g., maximum ring count ≤ 3 , maximum ring length ≤ 6) maintain near-baseline performance with FCD values around 1.07–1.08 [34], successfully removing extreme distribution tails while preserving common patterns. Loose constraints (e.g., maximum ring count ≤ 40 , maximum ring length ≤ 12) have virtually no impact since they lie beyond QM9’s natural range.

The practical implications are significant. For teams with expensive or frozen models, sampling-time projectors offer immediate structural control without retraining. Users can smoothly transition from highly constrained generation (yielding novel molecules) to unconstrained generation (yielding faithful reconstructions) by adjusting constraint parameters. This tunable trade-off provides direct control over the balance between specificity and diversity.

This thesis demonstrates that sampling-time projectors provide a powerful mechanism for constrained molecular generation. The approach works well, provides fine-grained control, and opens new possibilities for controllable generation in scenarios where model retraining is expensive or infeasible.

6.1 FUTURE WORK

Several promising directions emerge from this work. We could extend the framework to support additional constraint types, including chemical valence rules and functional-group restrictions. Developing methods to learn optimal edge ordering for constraint enforcement could improve solution quality under tight constraints. Another important direction is to explore how **multiple constraints** can be combined and jointly enforced during generation, for example simultaneously controlling both the total ring count and the maximum ring length. Such combinations would more closely reflect real chemical design rules but also introduce challenges when constraints interact or conflict. Extending evaluation to larger, more diverse molecular datasets would help us understand the general applicability of sampling-time constraint enforcement.

A particularly promising direction involves extending our framework to enforce chemical property constraints, such as **Lipinski’s Rule of Five** [28]. While our current work focused on structural constraints (ring counts and lengths), property-based constraints would require more sophisticated projectors that can evaluate molecular properties like molecular weight, logP, hydrogen-bond donors/acceptors, and topological polar surface area. This represents a natural progression from structural control to chemical-property control and would significantly expand the practical applicability of our approach for drug-discovery applications. The challenge lies in developing efficient projectors that can evaluate these properties during the sampling process, potentially requiring cached property calculations or approximate methods that balance accuracy with computational efficiency.

The broader implications of this work extend beyond molecular generation. Our findings suggest that sampling-time projectors could be valuable for any graph-generation task where structural constraints are important. The key insight is that there exists a fundamental trade-off between constraint strictness and distributional fidelity, and this trade-off can be controlled precisely through constraint parameters. This makes these projectors a powerful tool for controllable generation, particularly in scenarios where model retraining is expensive or infeasible.

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